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(12) **United States Patent**
Krastins et al.

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(54) **SURGICAL INSTRUMENTS AND METHODS
FOR PERFORMING TONSILLECTOMY,
ADENOIDECTOMY, AND OTHER
SURGICAL PROCEDURES**

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(73) Assignee: **Covidien LP**, Mansfield, MA (US)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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(22) Filed: **Jul. 29, 2023**

(65) **Prior Publication Data**

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Related U.S. Application Data

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Oct. 7, 2020, now Pat. No. 11,730,532, which is a
(Continued)

(51) **Int. Cl.**

A61B 18/14 (2006.01)

A61B 17/26 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **A61B 18/12** (2013.01); **A61B 17/26**
(2013.01); **A61B 17/29** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC A61B 17/0218; A61B 17/26; A61B 17/29;
A61B 17/295; A61B 18/12;

(Continued)

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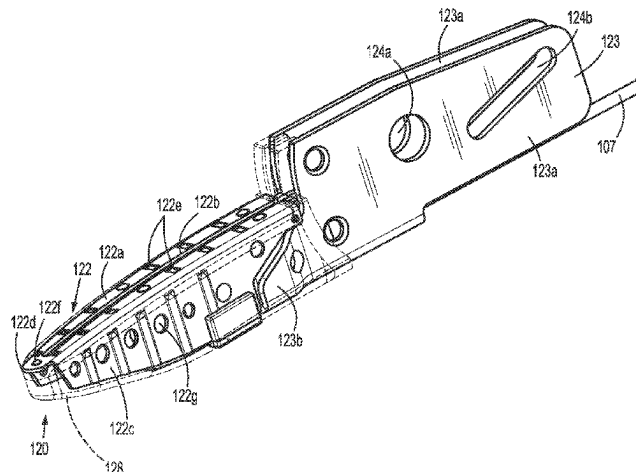
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Primary Examiner — Michael F Peffley

(57) **ABSTRACT**

A surgical instrument includes a housing, a shaft extending therefrom, an end effector assembly supported by the shaft, a movable handle, and a drive assembly. The drive assembly includes a translatable drive member for actuating the end effector assembly, and a torsion spring including first and second legs. The first leg is configured to translate through the housing in response to movement of the movable handle relative to the housing. The second leg is configured to translate through the housing in cooperation with the first leg to move the drive member longitudinally when a force acting on the drive member is less than a threshold force, and to remain in fixed position, thereby tensioning the torsion spring and retaining the drive member in fixed position when the force acting on the drive member is equal to or exceeds the threshold force.

20 Claims, 29 Drawing Sheets



continuation of application No. 15/806,679, filed on Nov. 8, 2017, now Pat. No. 10,828,084, which is a continuation of application No. 14/719,422, filed on May 22, 2015, now Pat. No. 9,918,779.

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| <i>A61B 17/29</i> | (2006.01) |
| <i>A61B 18/12</i> | (2006.01) |
| <i>A61B 90/00</i> | (2016.01) |
| <i>A61B 17/00</i> | (2006.01) |
| <i>A61B 17/02</i> | (2006.01) |
| <i>A61B 17/295</i> | (2006.01) |
| <i>A61B 17/32</i> | (2006.01) |
| <i>A61B 18/00</i> | (2006.01) |

- CPC ***A61B 18/1445*** (2013.01); ***A61B 18/1485***
(2013.01); ***A61B 90/03*** (2016.02); ***A61B***
2017/00353 (2013.01); ***A61B 2017/00424***
(2013.01); ***A61B 2017/00526*** (2013.01); ***A61B***
17/0218 (2013.01); ***A61B 2017/2912***
(2013.01); ***A61B 2017/2919*** (2013.01); ***A61B***
2017/292 (2013.01); ***A61B 2017/2925***
(2013.01); ***A61B 2017/2926*** (2013.01); ***A61B***
2017/2936 (2013.01); ***A61B 17/295*** (2013.01);
A61B 2017/320044 (2013.01); ***A61B***
2018/00184 (2013.01); ***A61B 2018/00327***
(2013.01); ***A61B 2018/00607*** (2013.01); ***A61B***
2018/0091 (2013.01); ***A61B 2018/00928***
(2013.01); ***A61B 18/1447*** (2013.01); ***A61B***
2018/1455 (2013.01)

- CPC A61B 18/1206; A61B 18/1445; A61B
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2017/00184; A61B 2017/00327; A61B
2017/00589; A61B 2017/00595; A61B
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See application file for complete search history.

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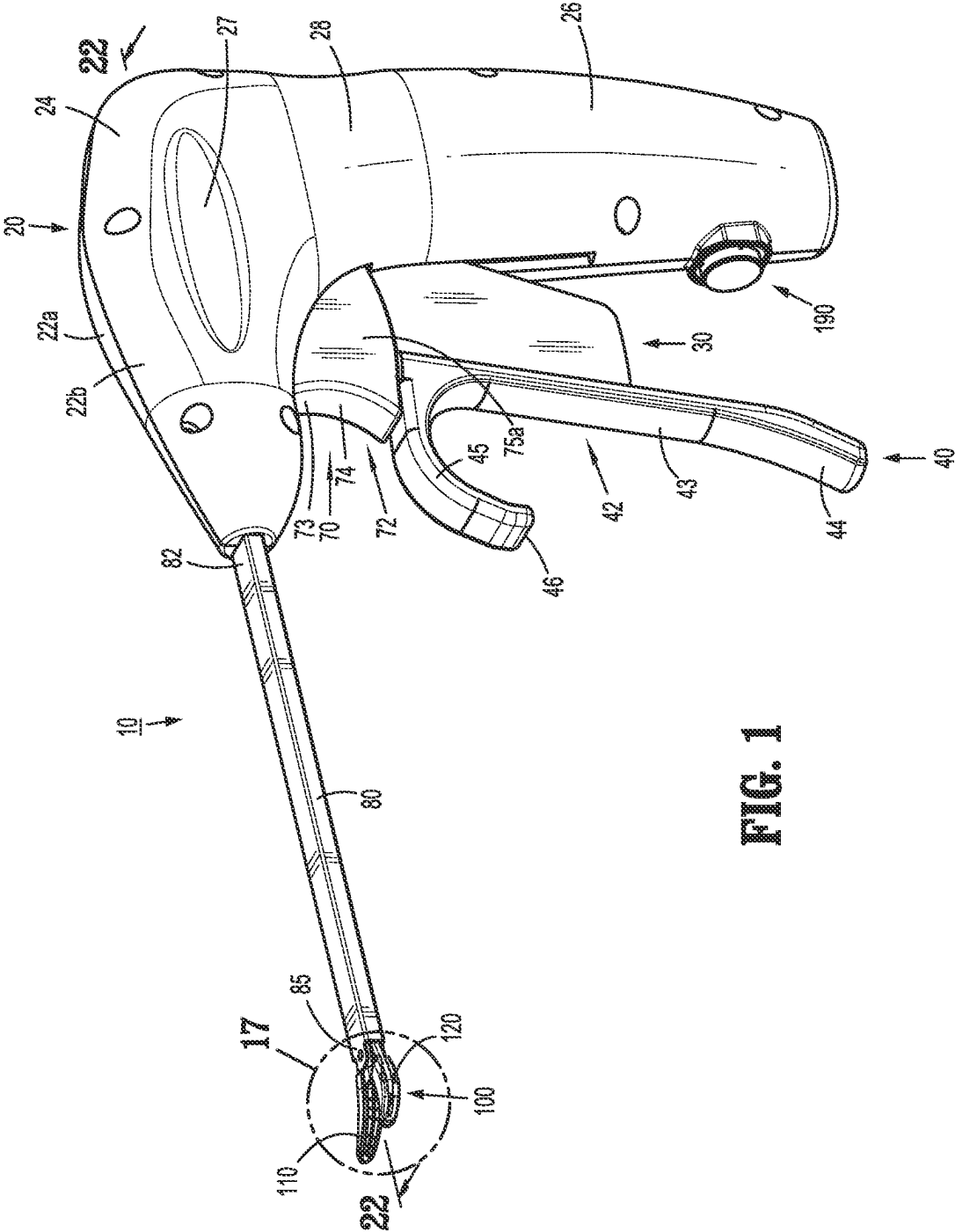
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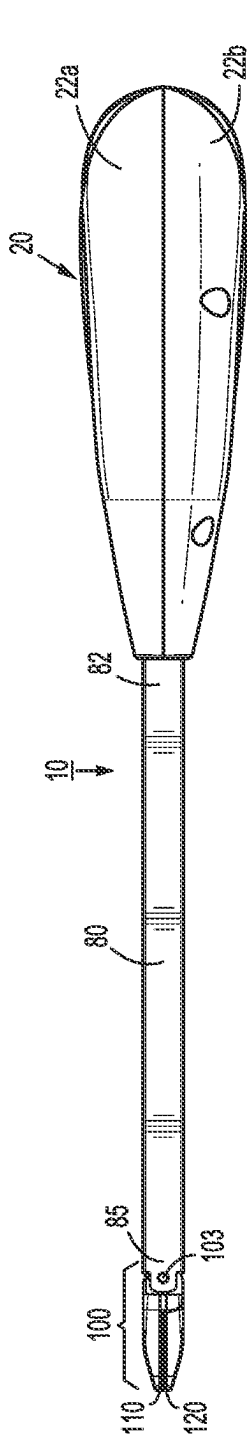


FIG. 2

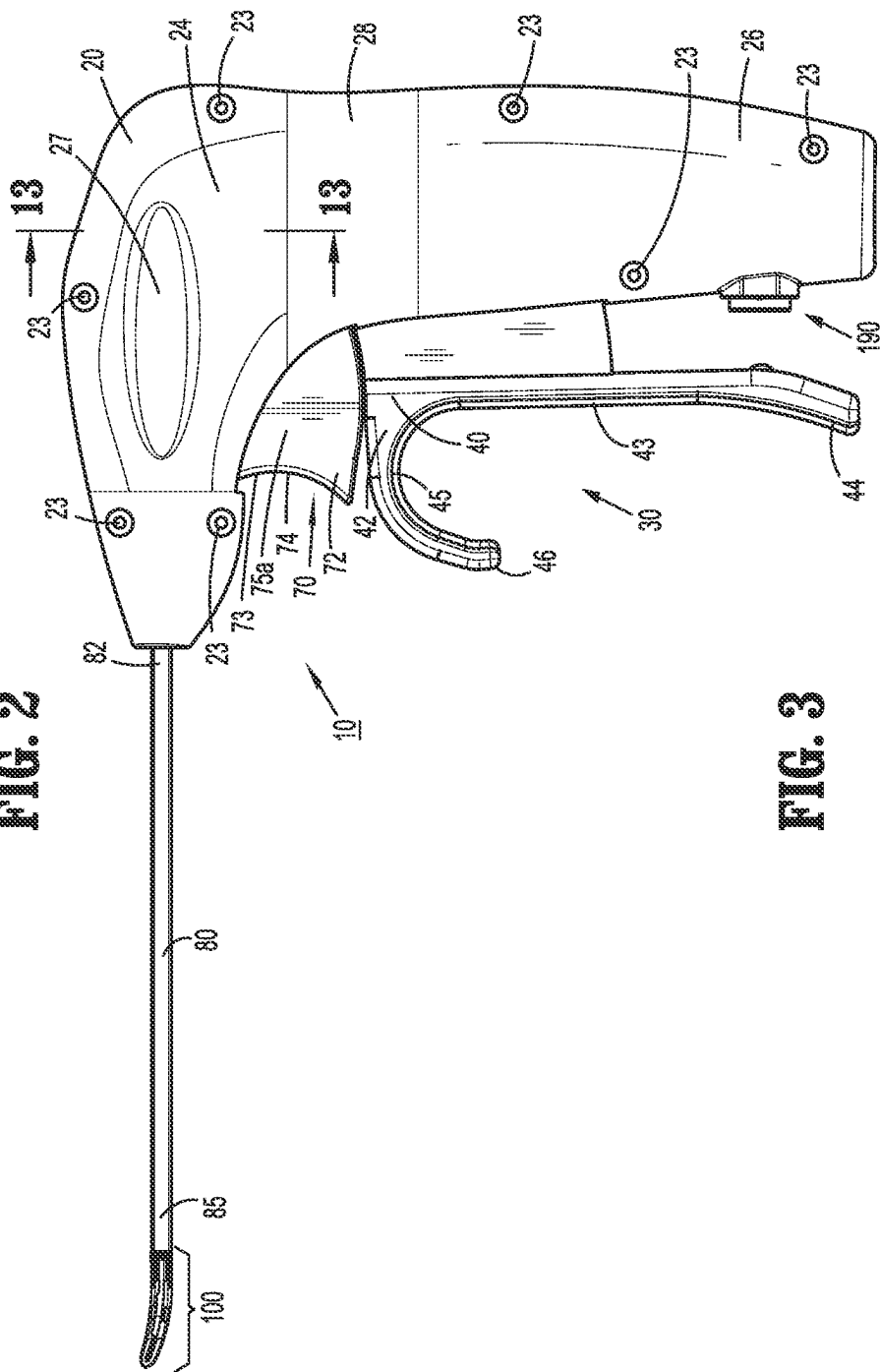
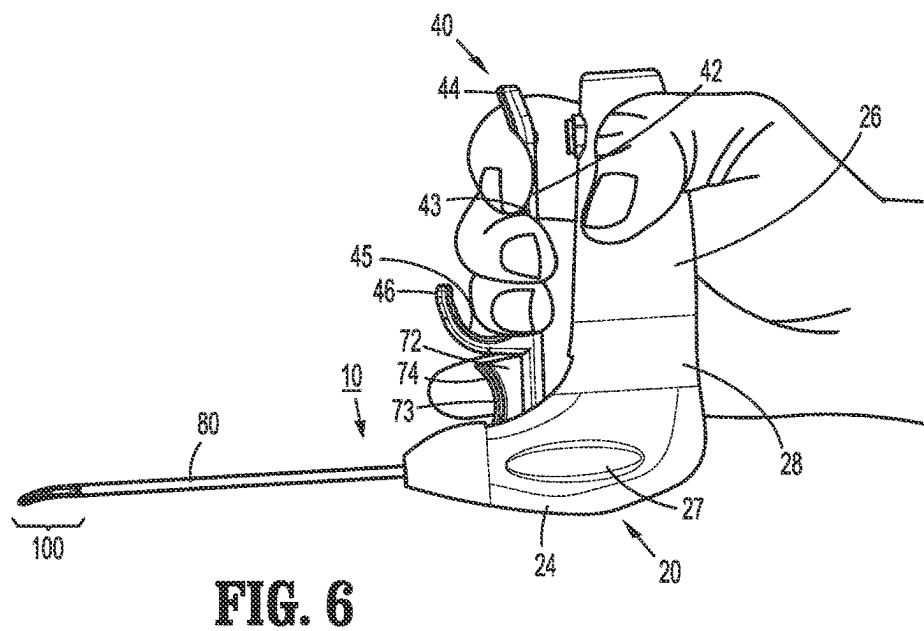
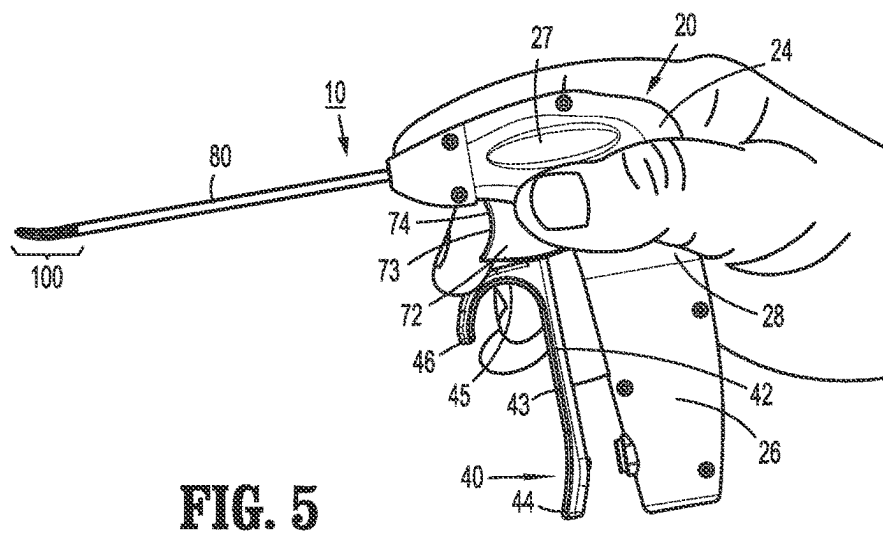
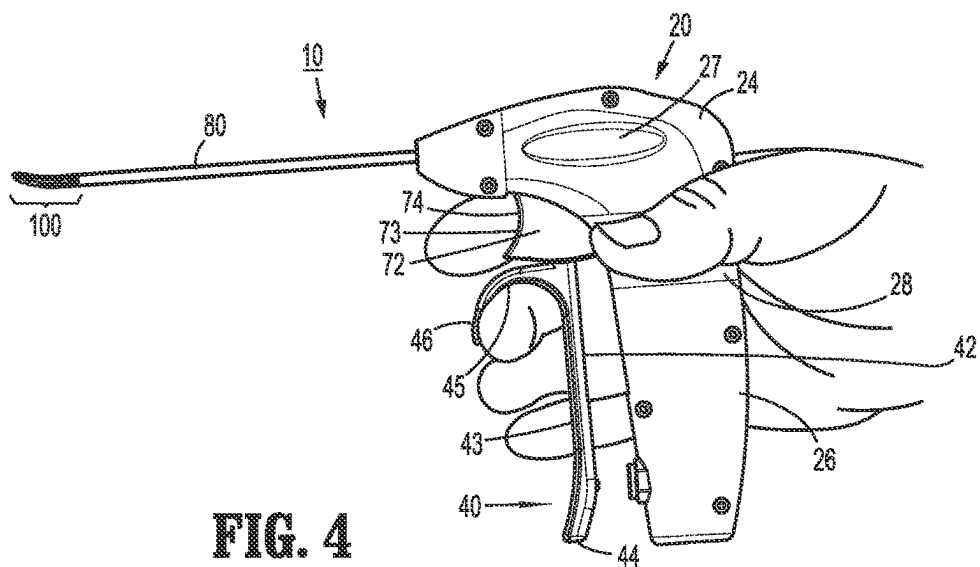
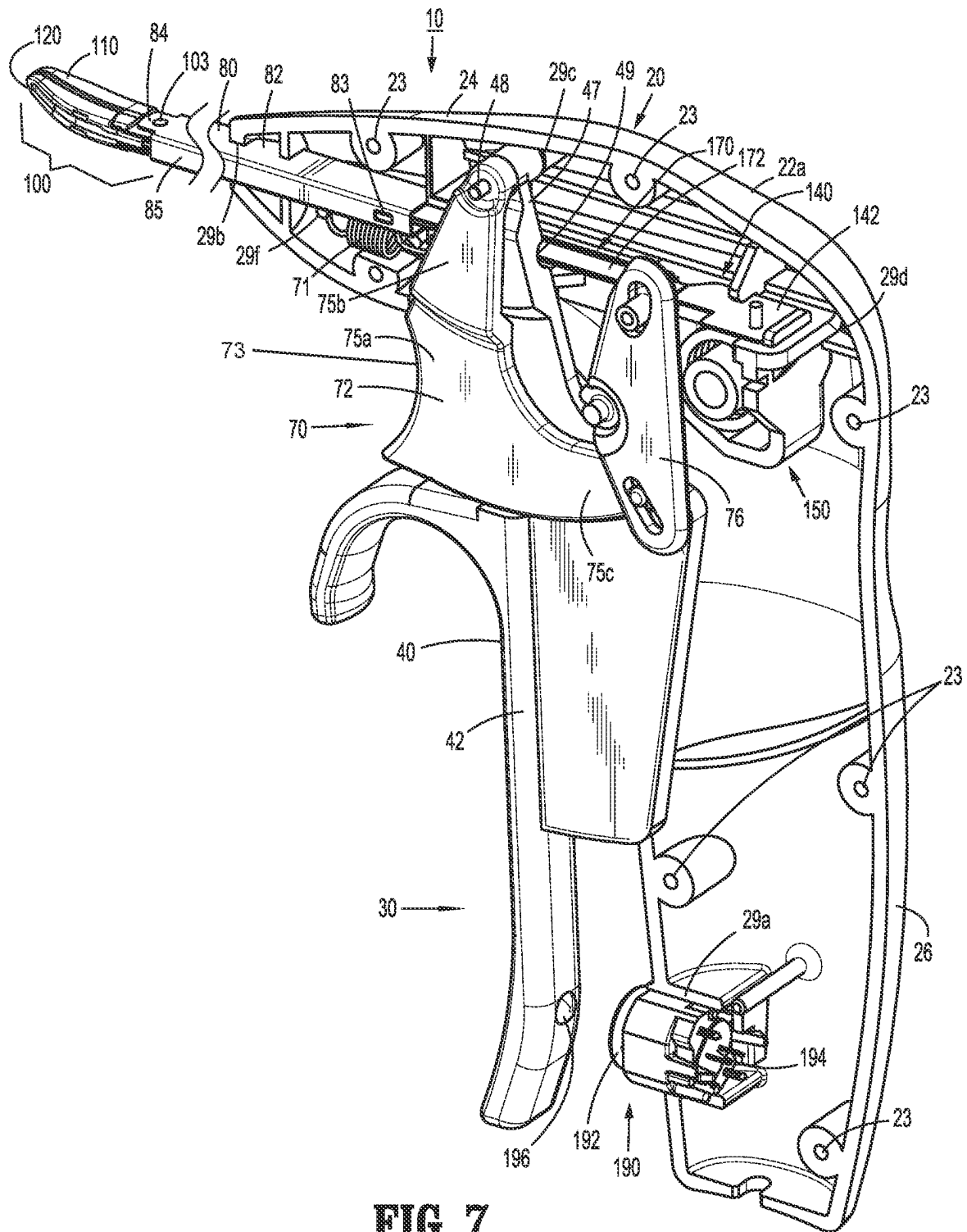
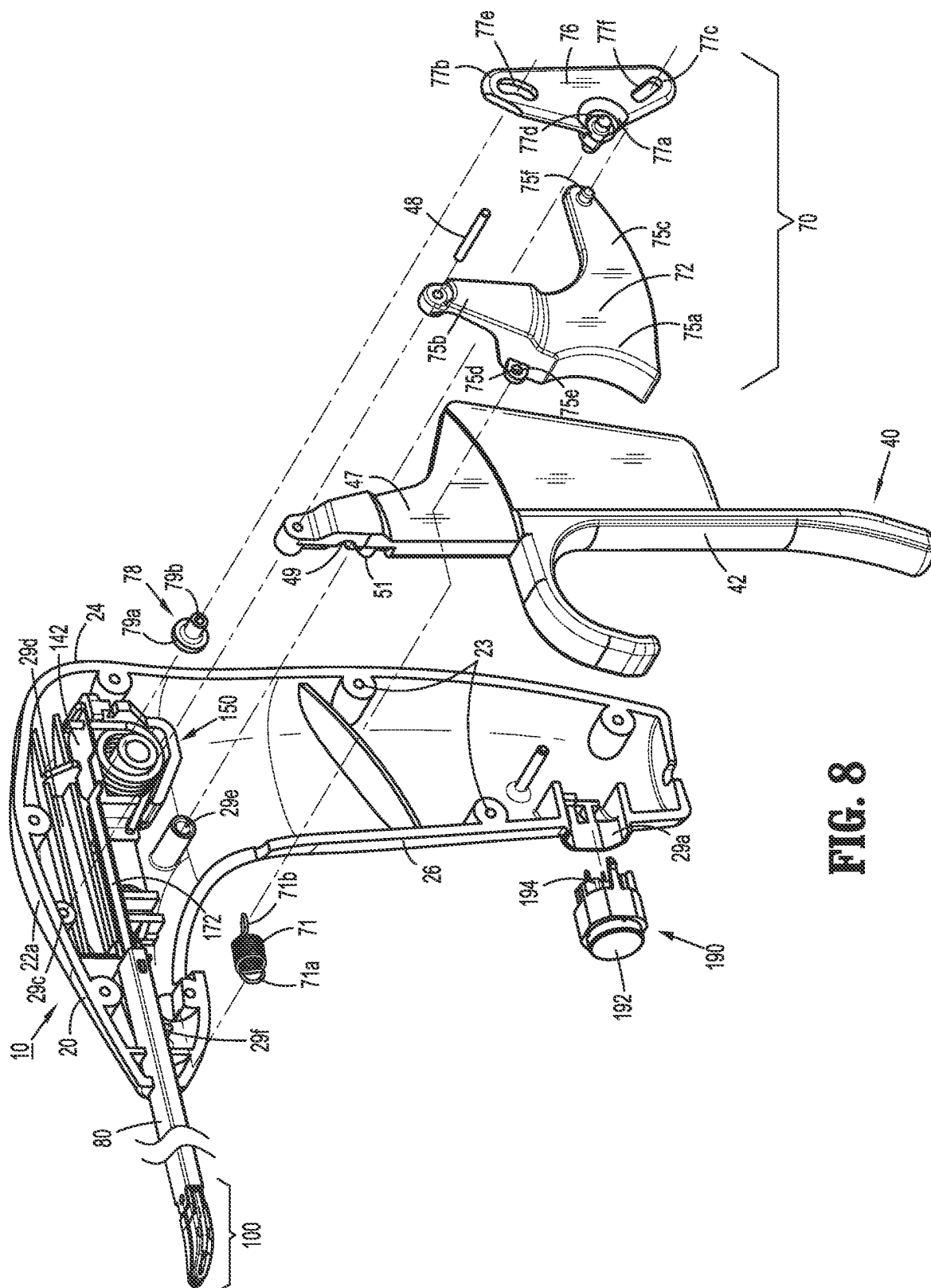


FIG. 3







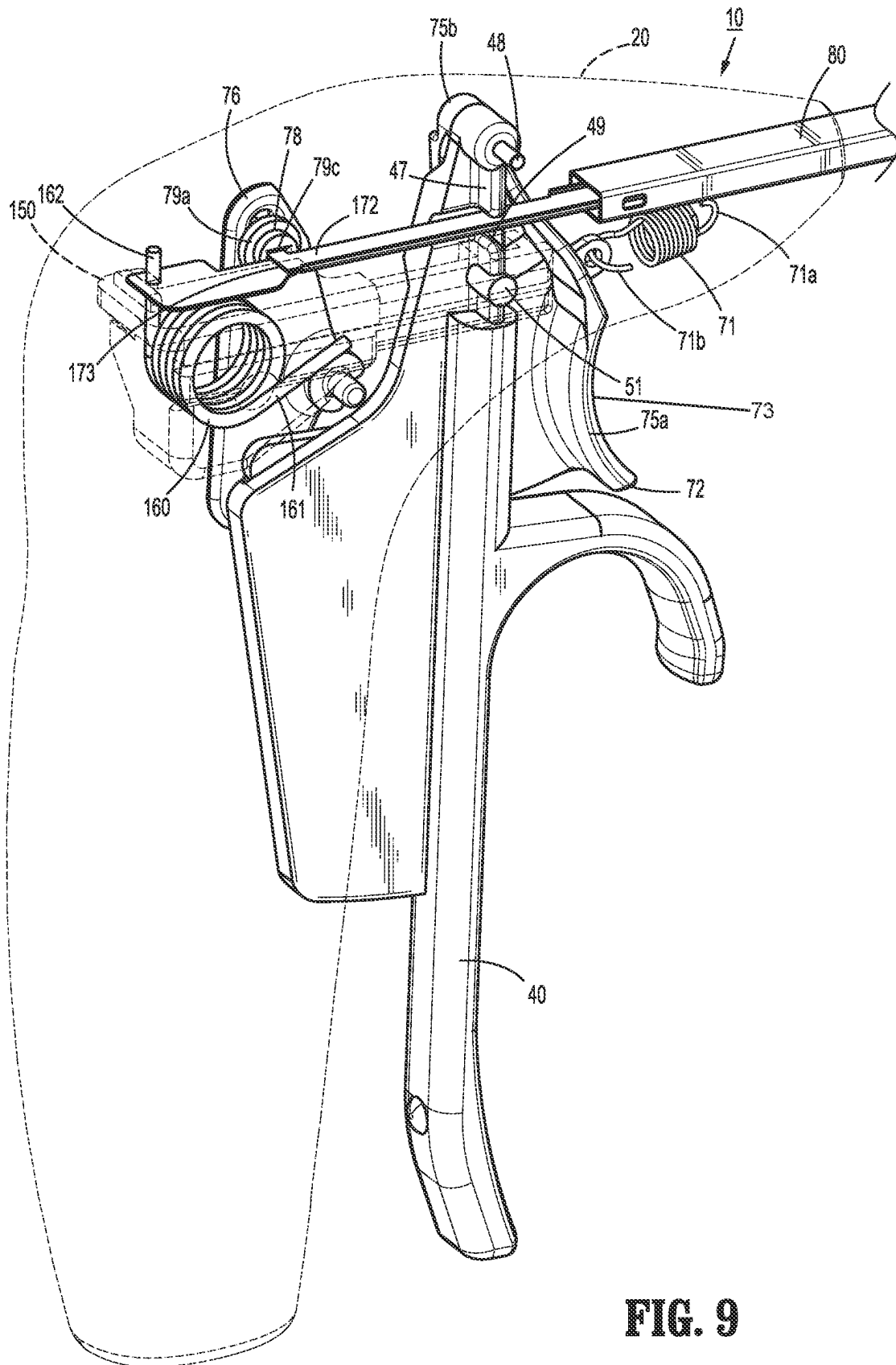
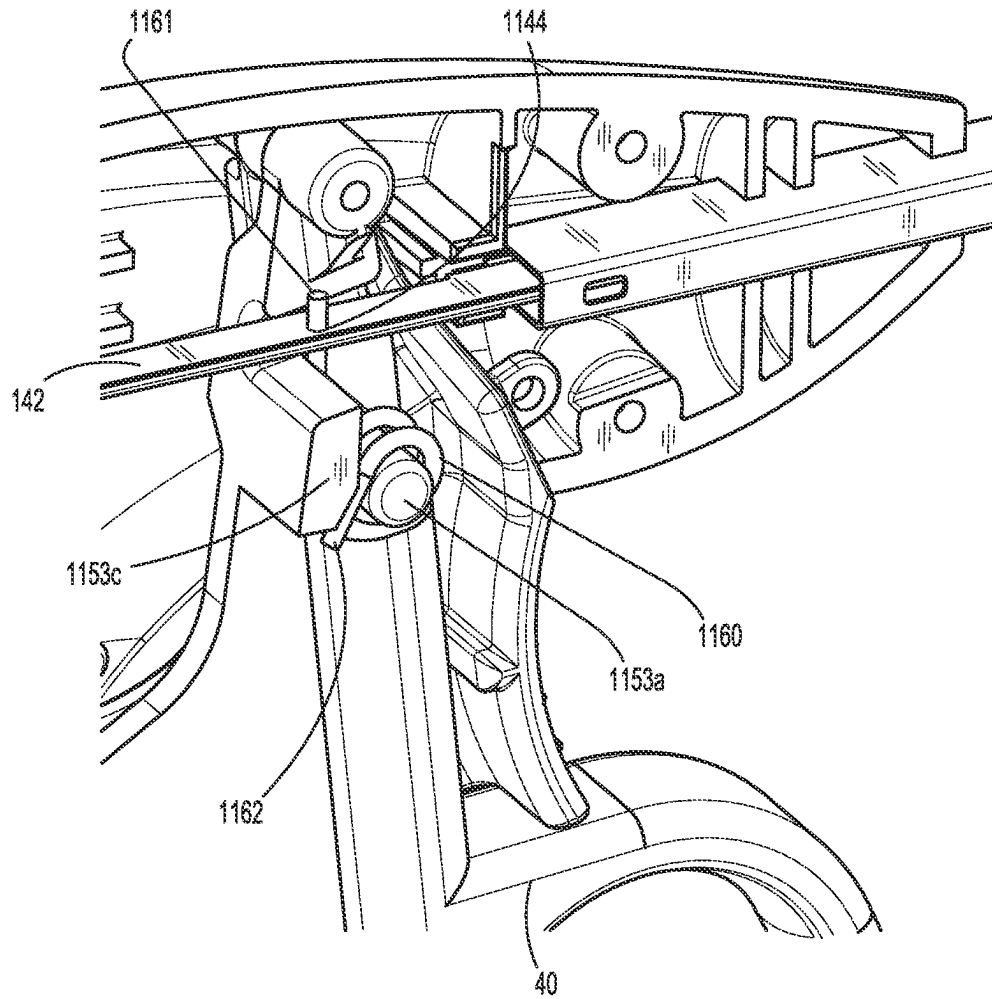


FIG. 9

**FIG. 9A**

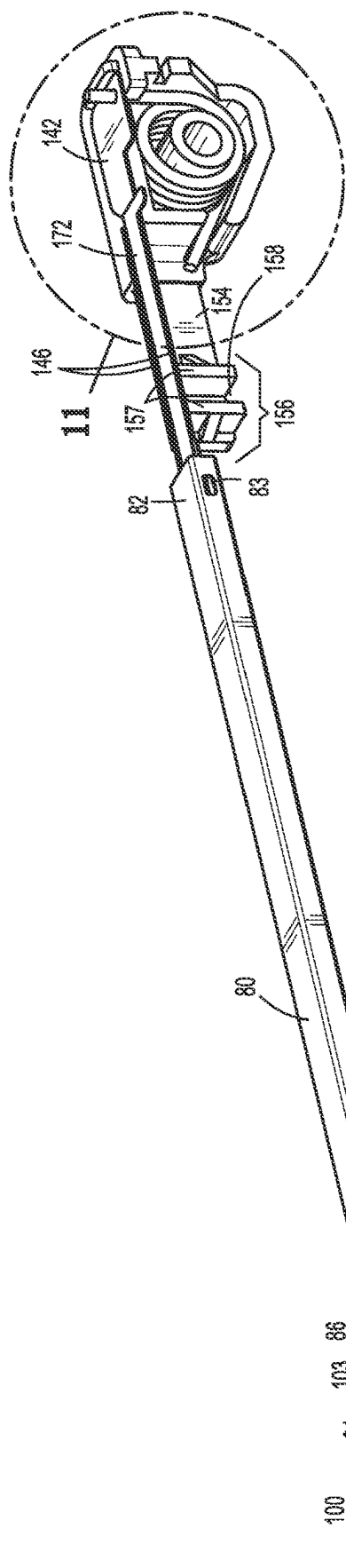


FIG. 10

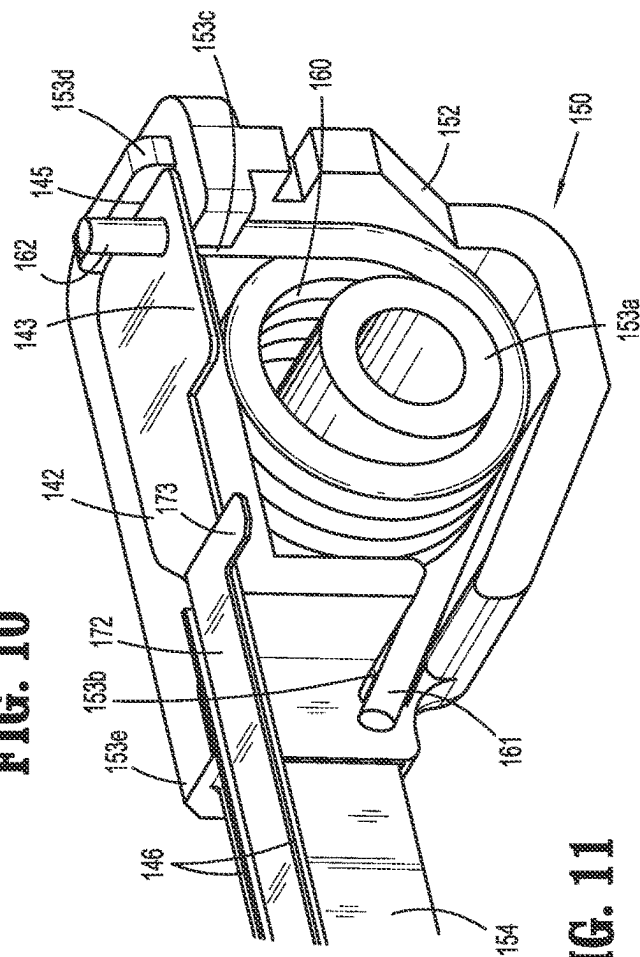


FIG. 11

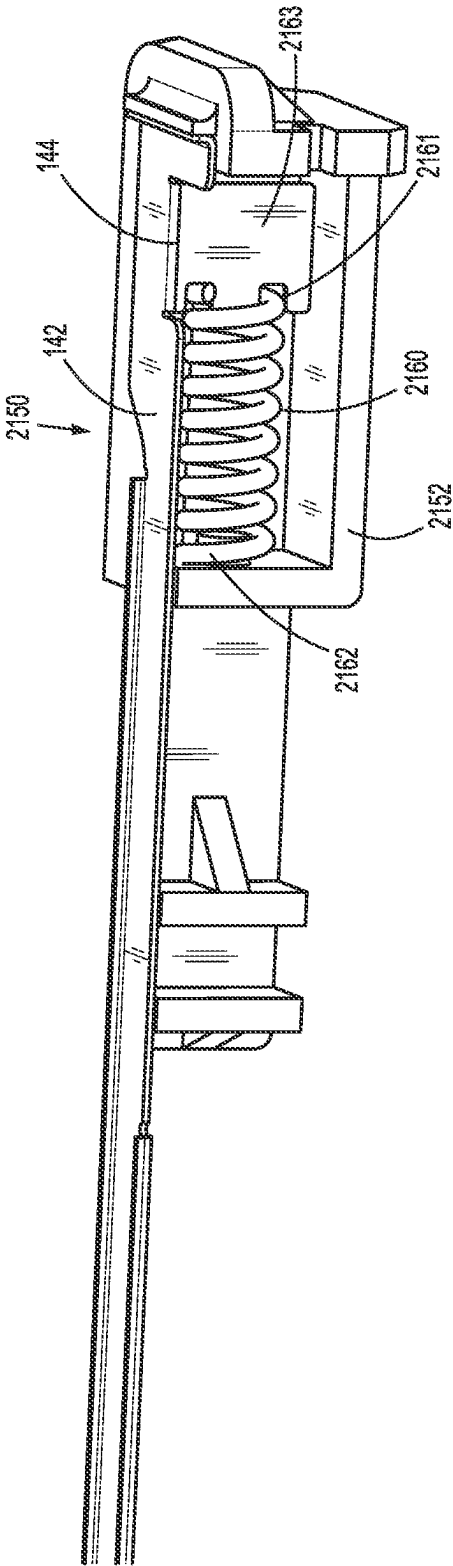
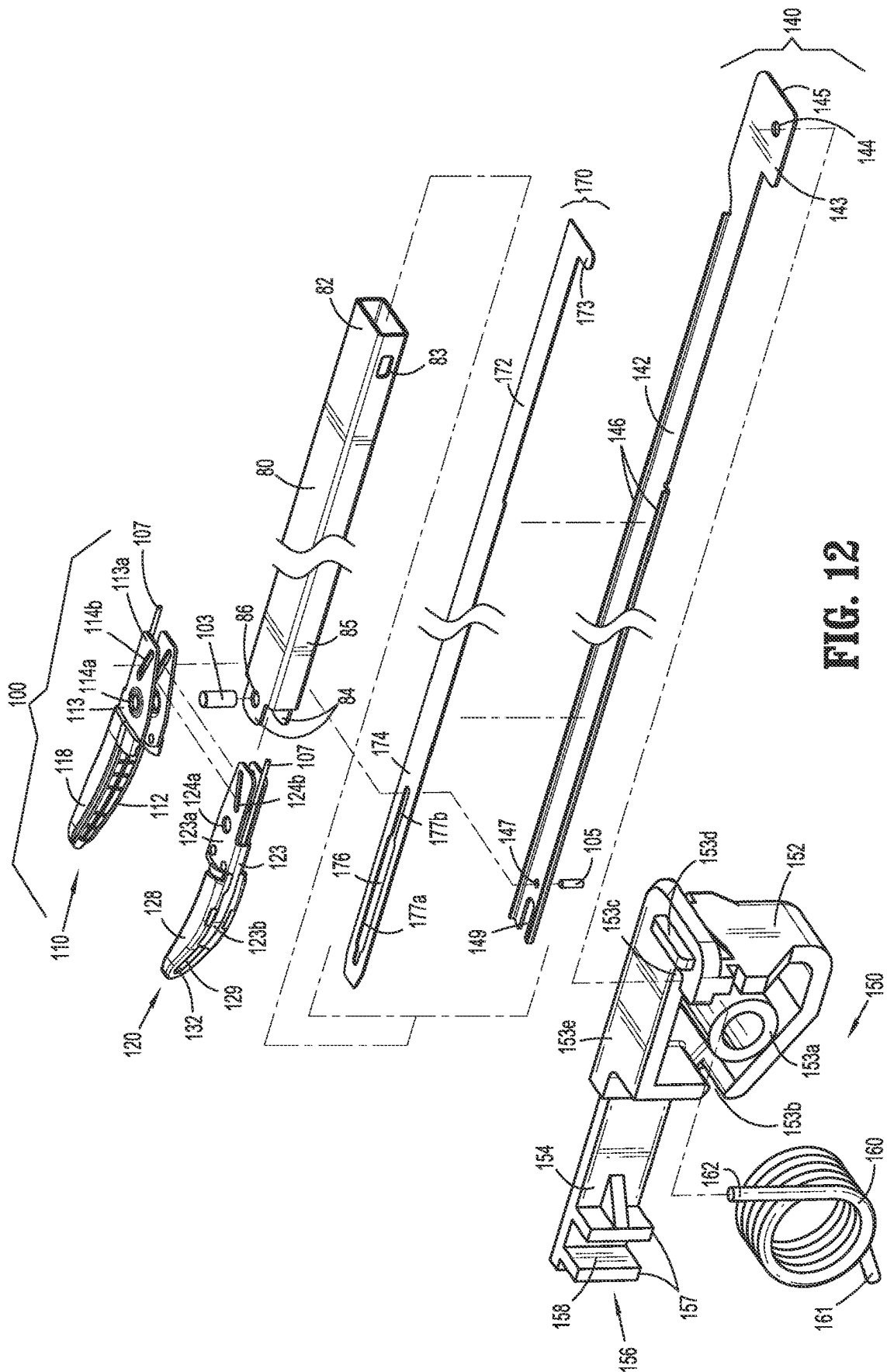


FIG. 11A



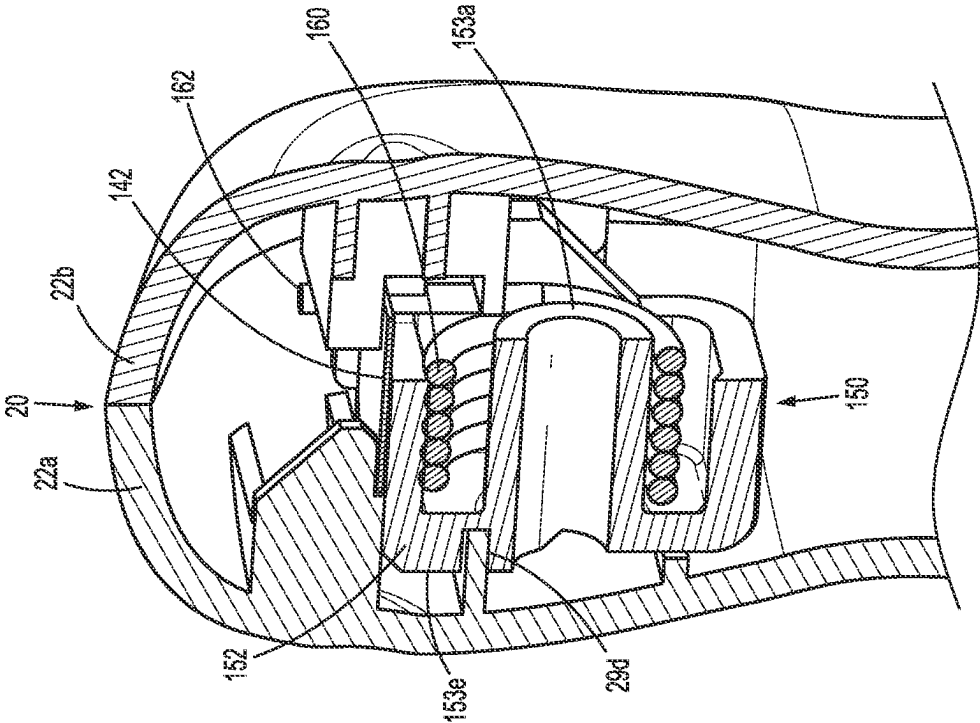


FIG. 13

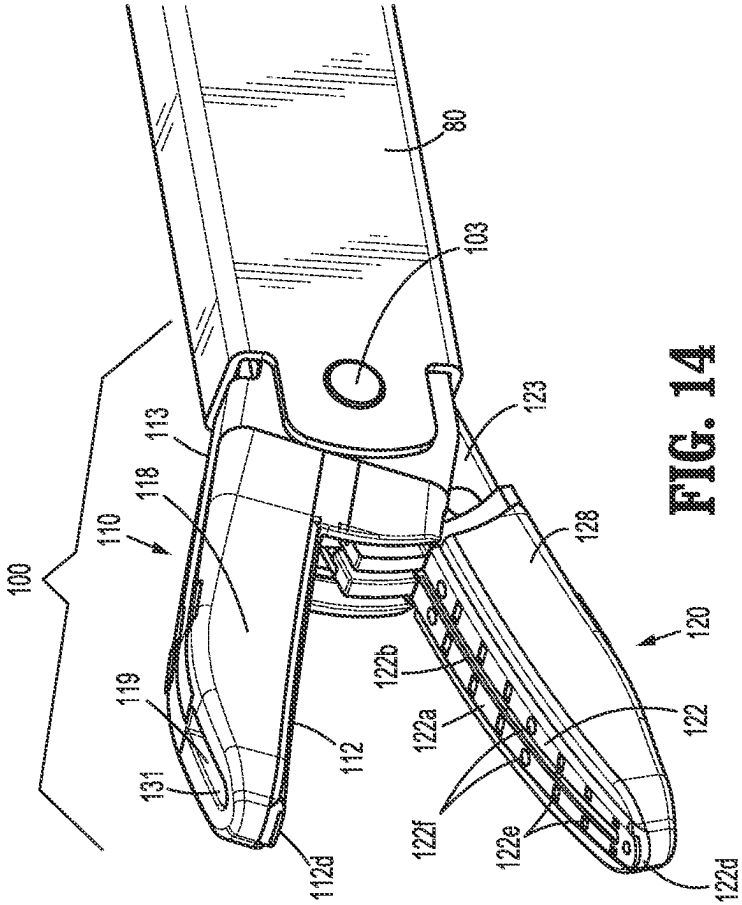


FIG. 14

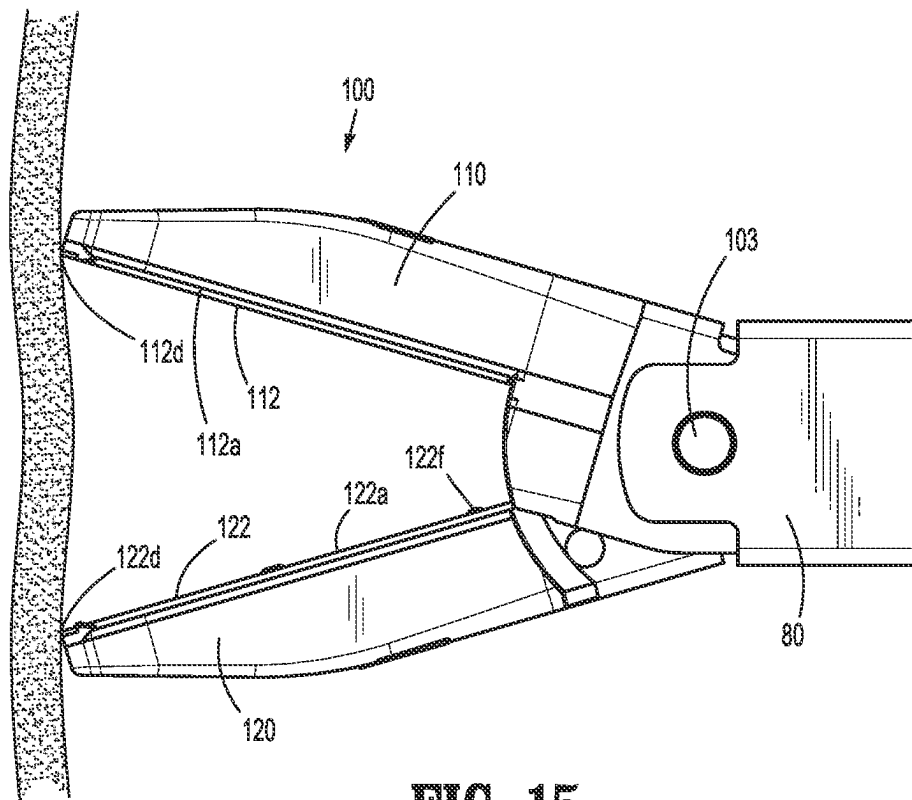


FIG. 15

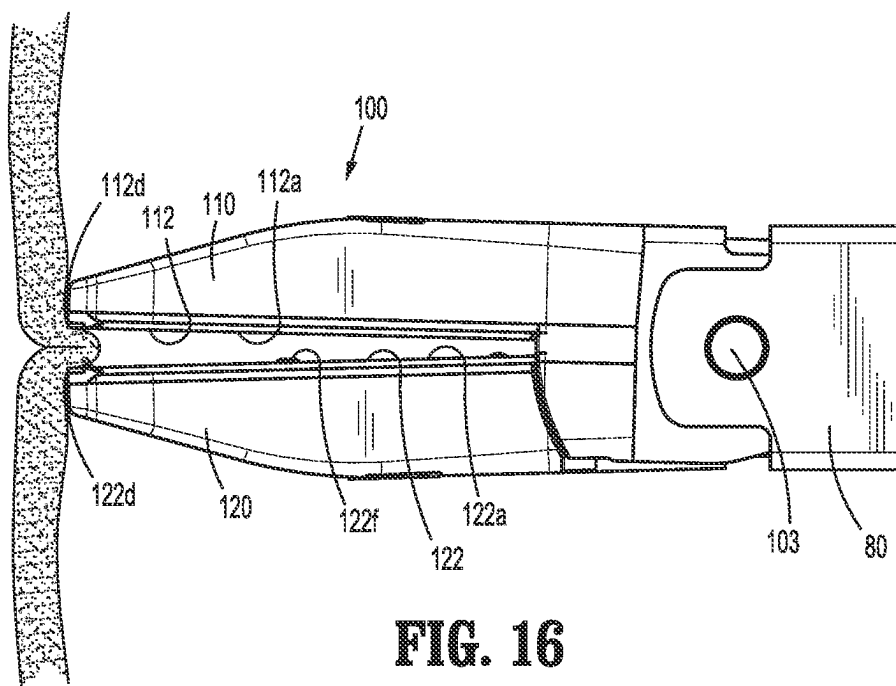


FIG. 16

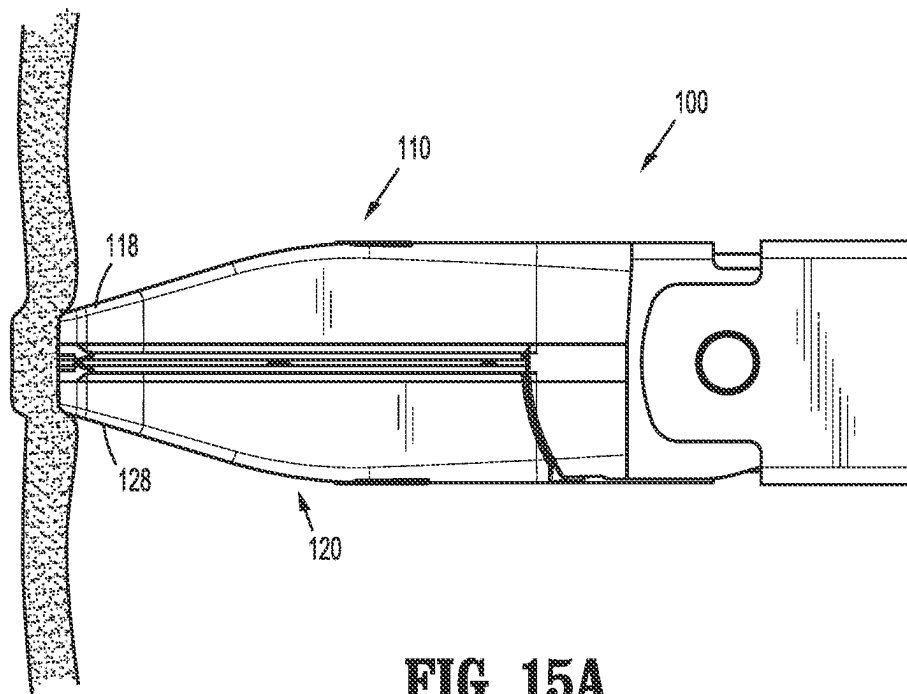


FIG. 15A

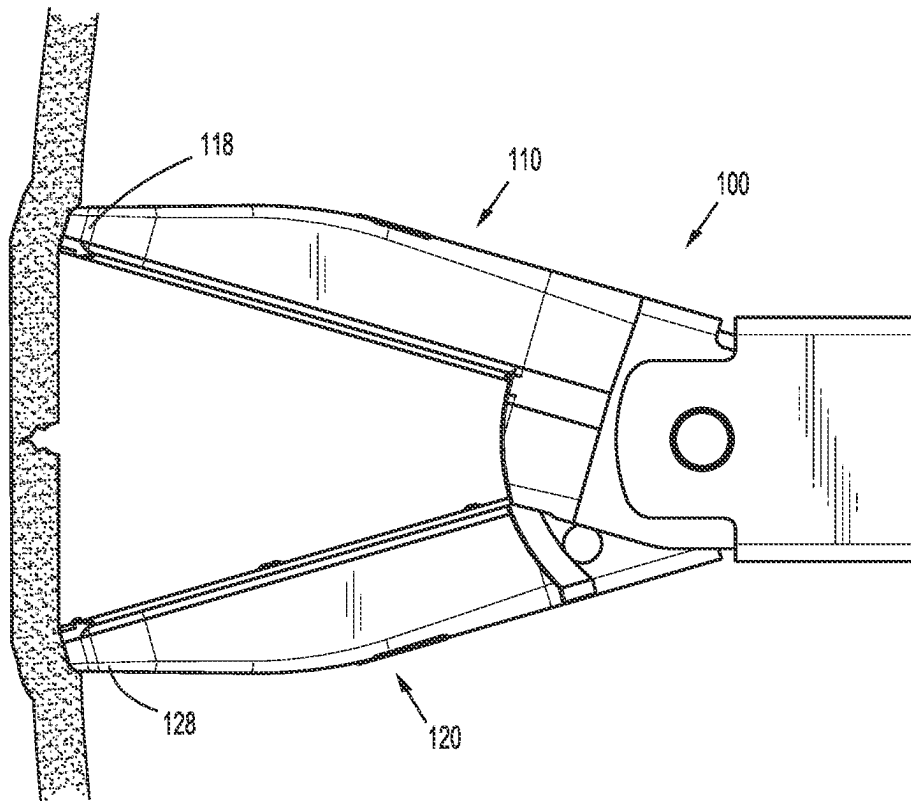
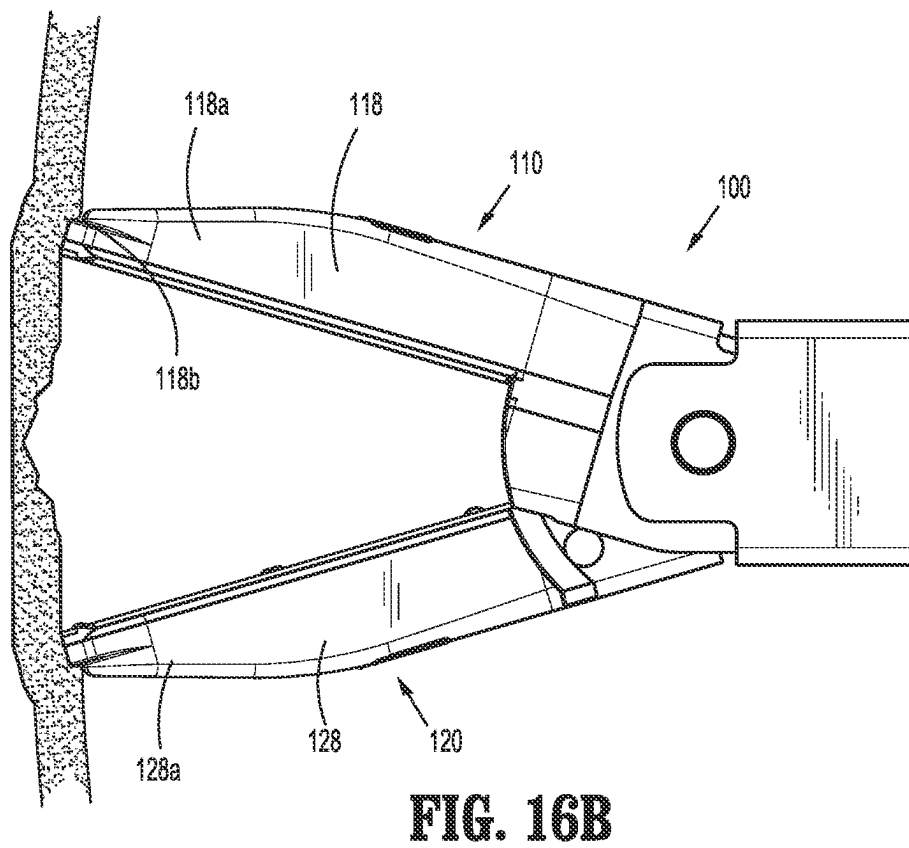
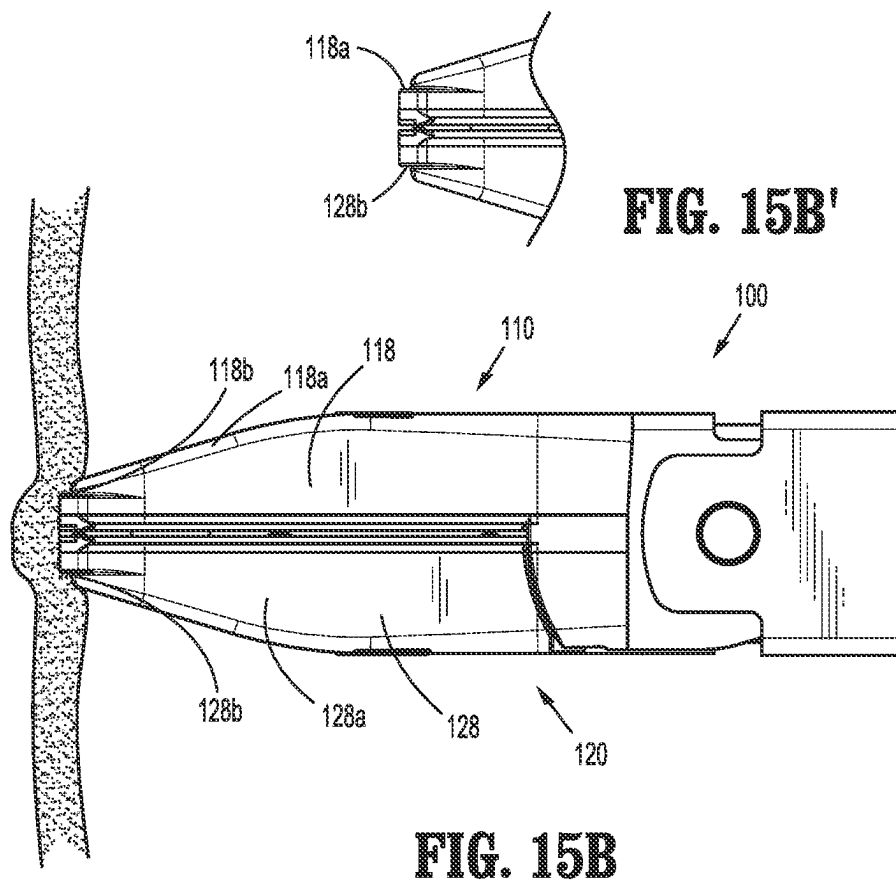


FIG. 16A



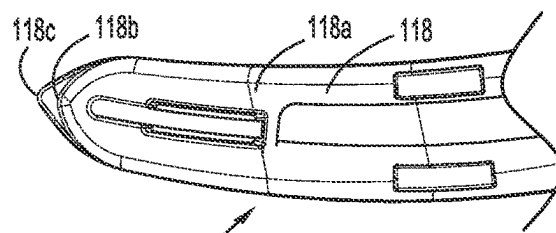


FIG. 15C'

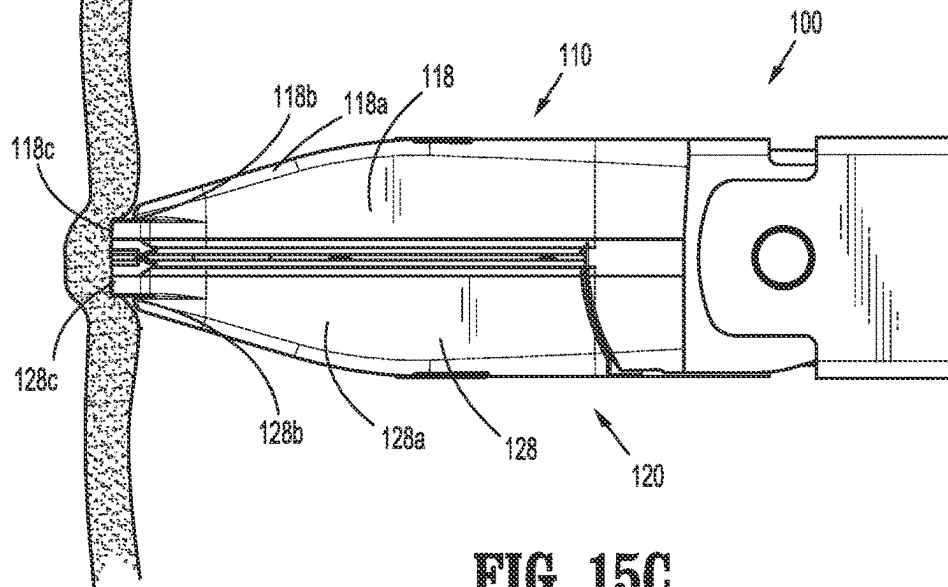


FIG. 15C

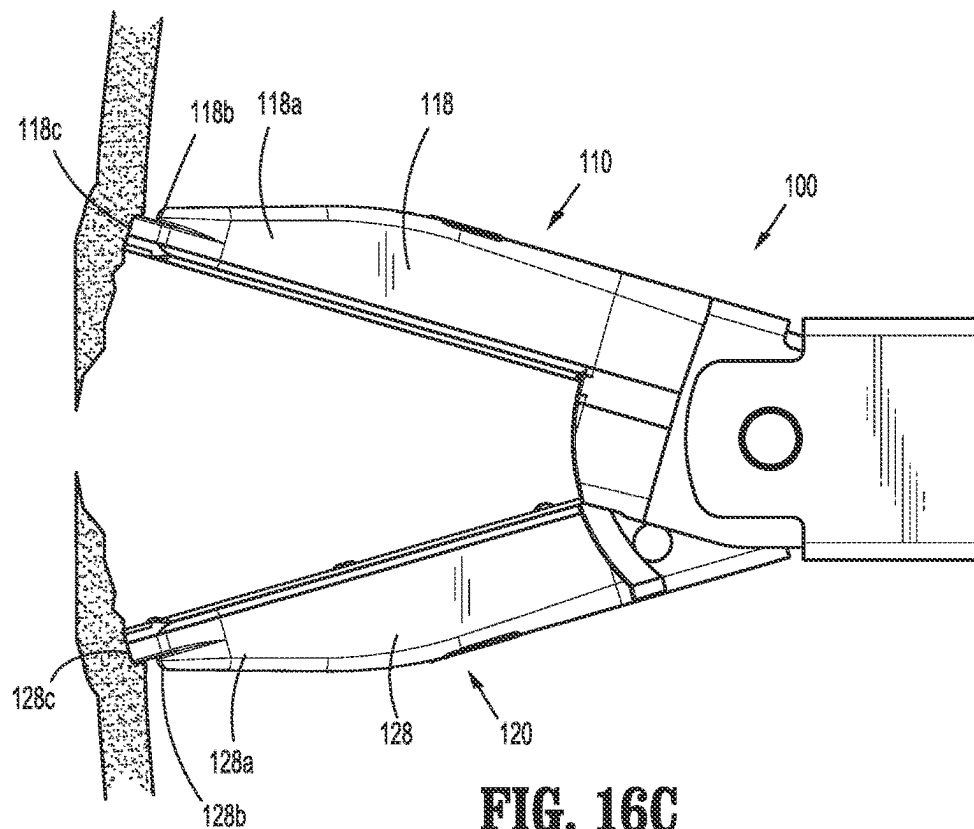
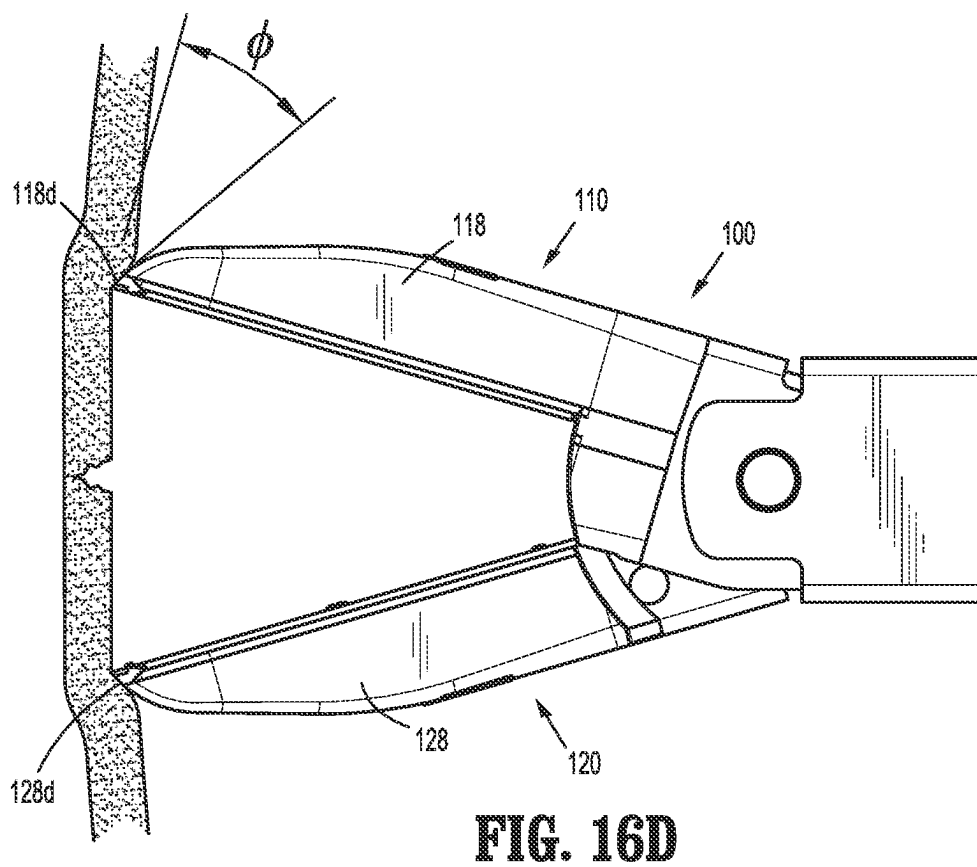
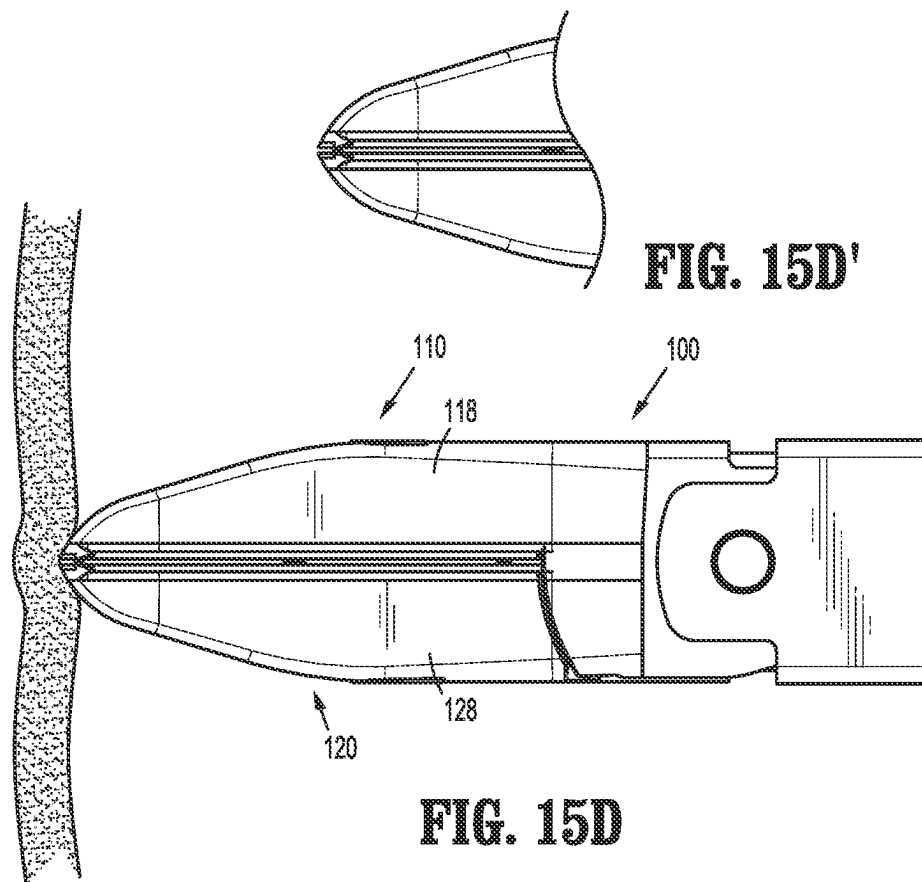
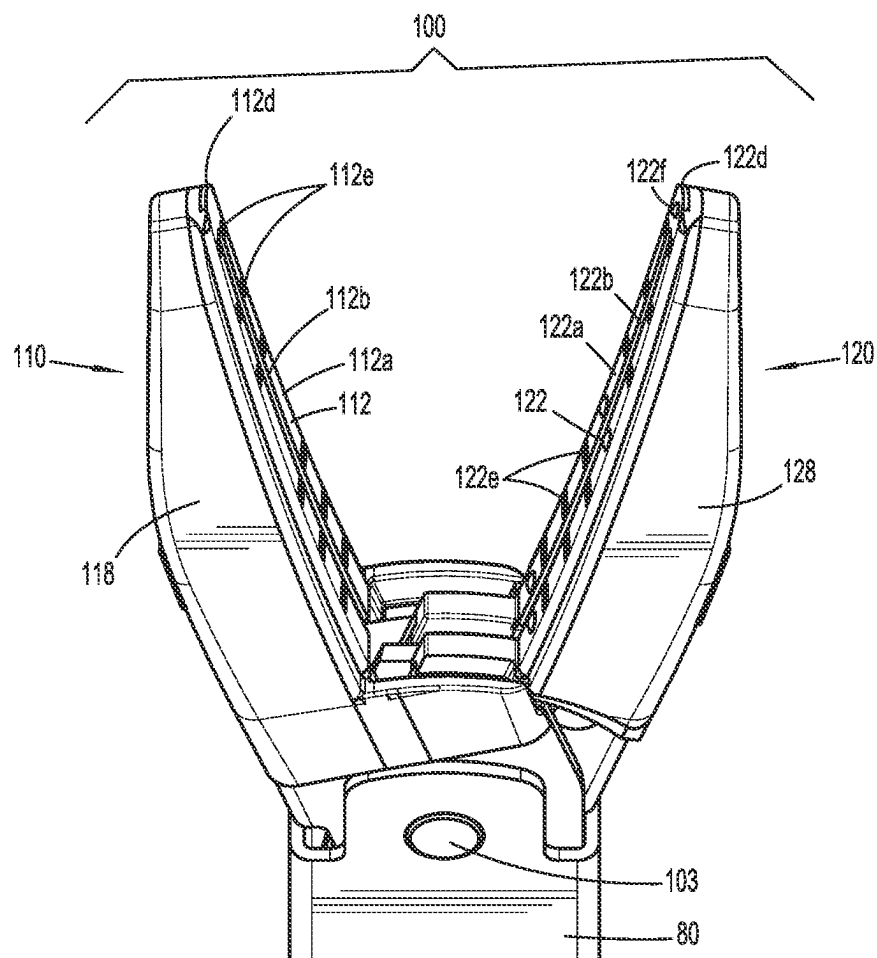
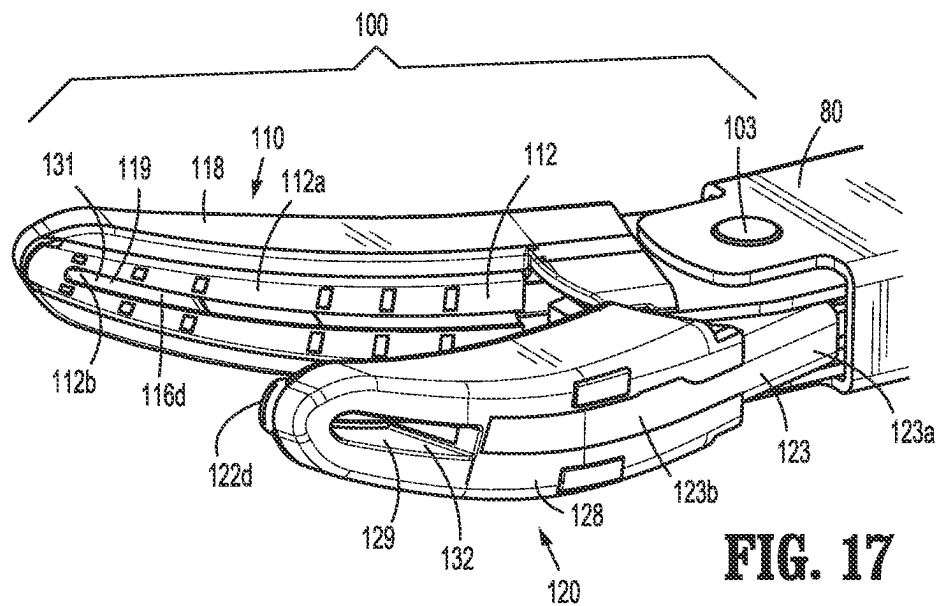


FIG. 16C





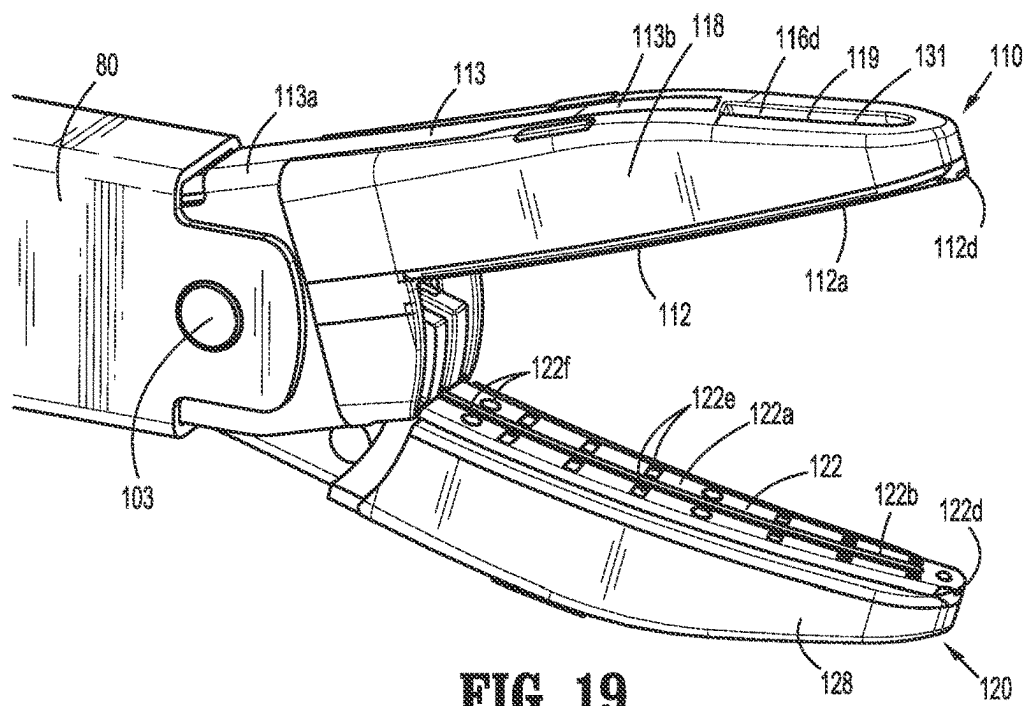


FIG. 19

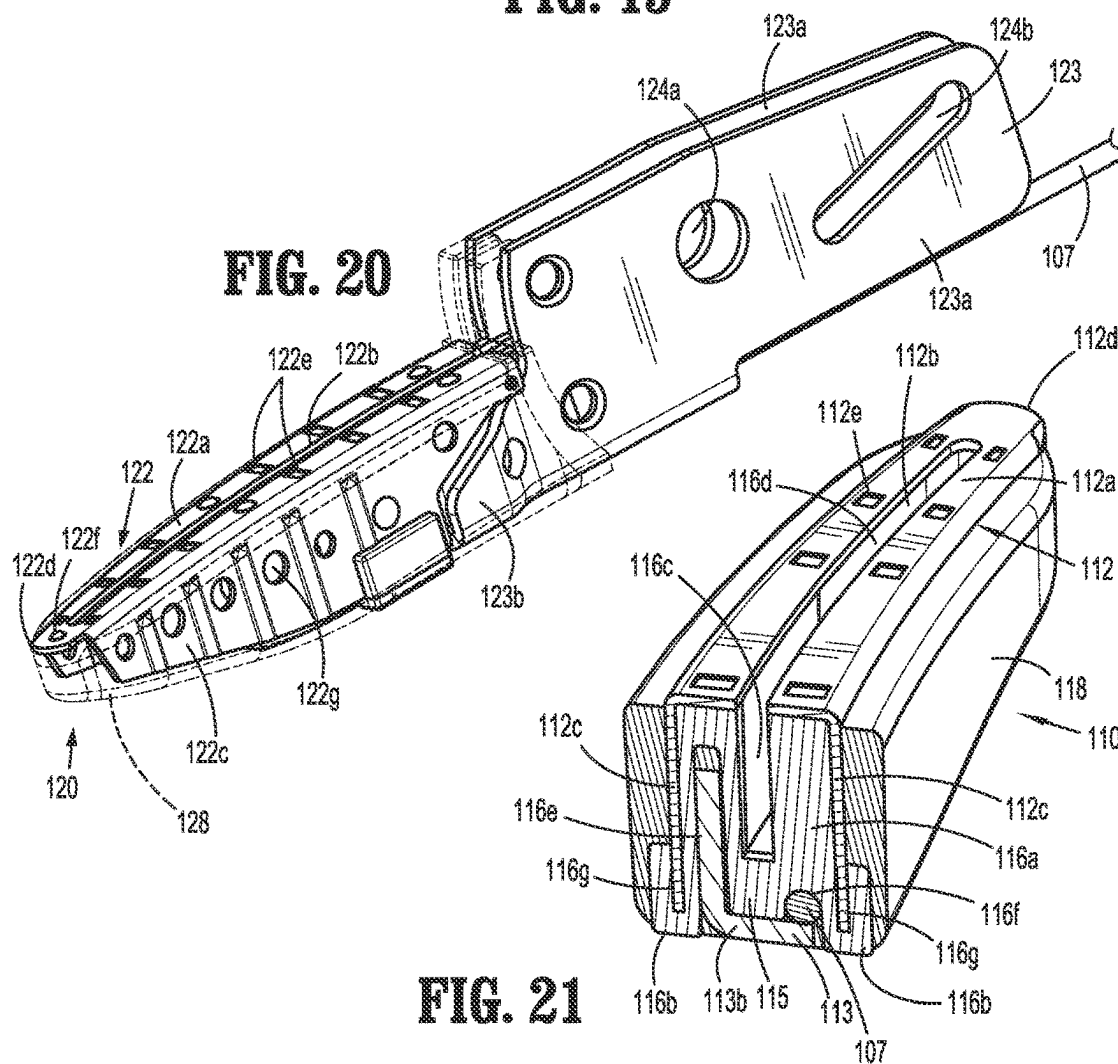


FIG. 20

FIG. 21

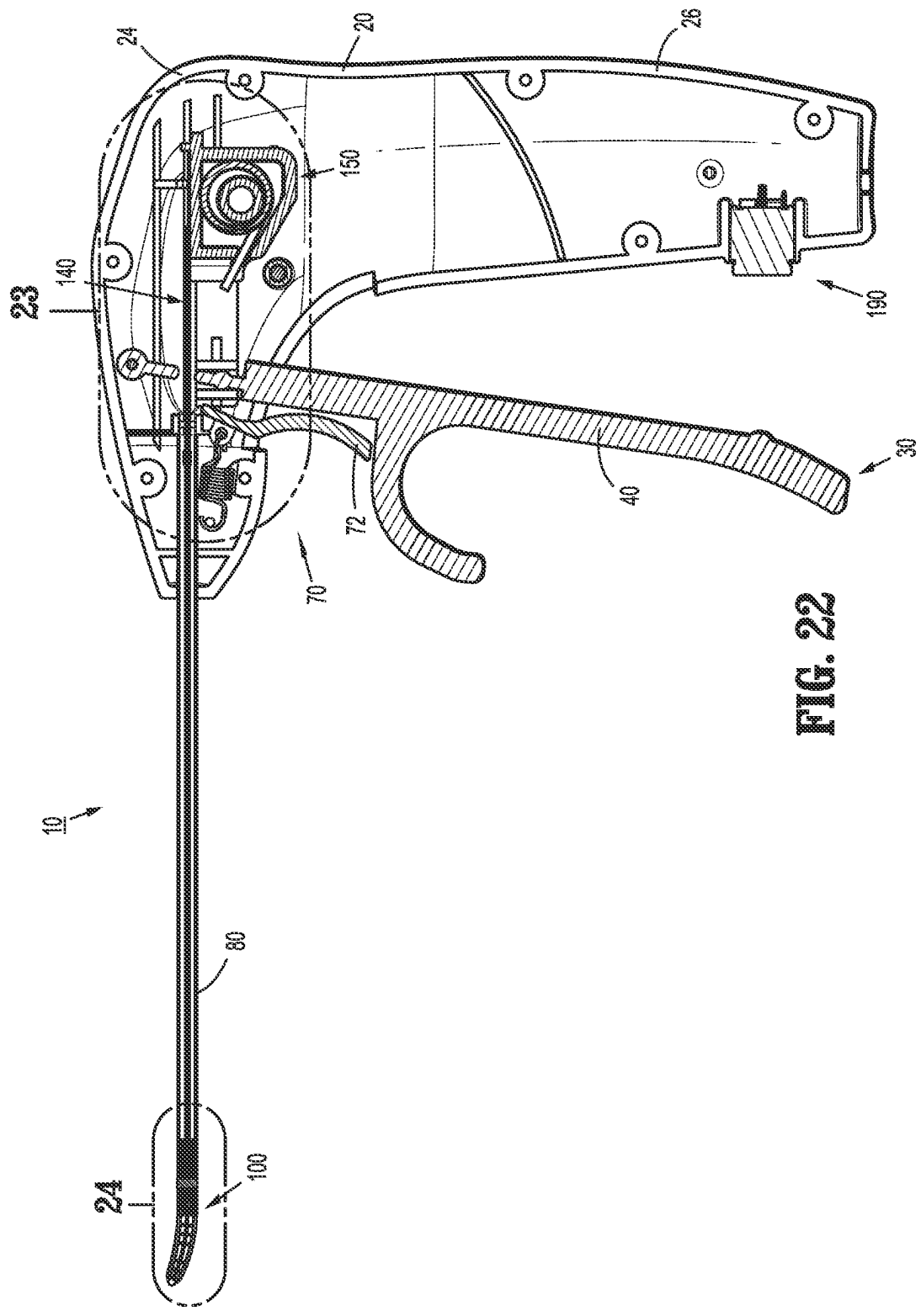


FIG. 22

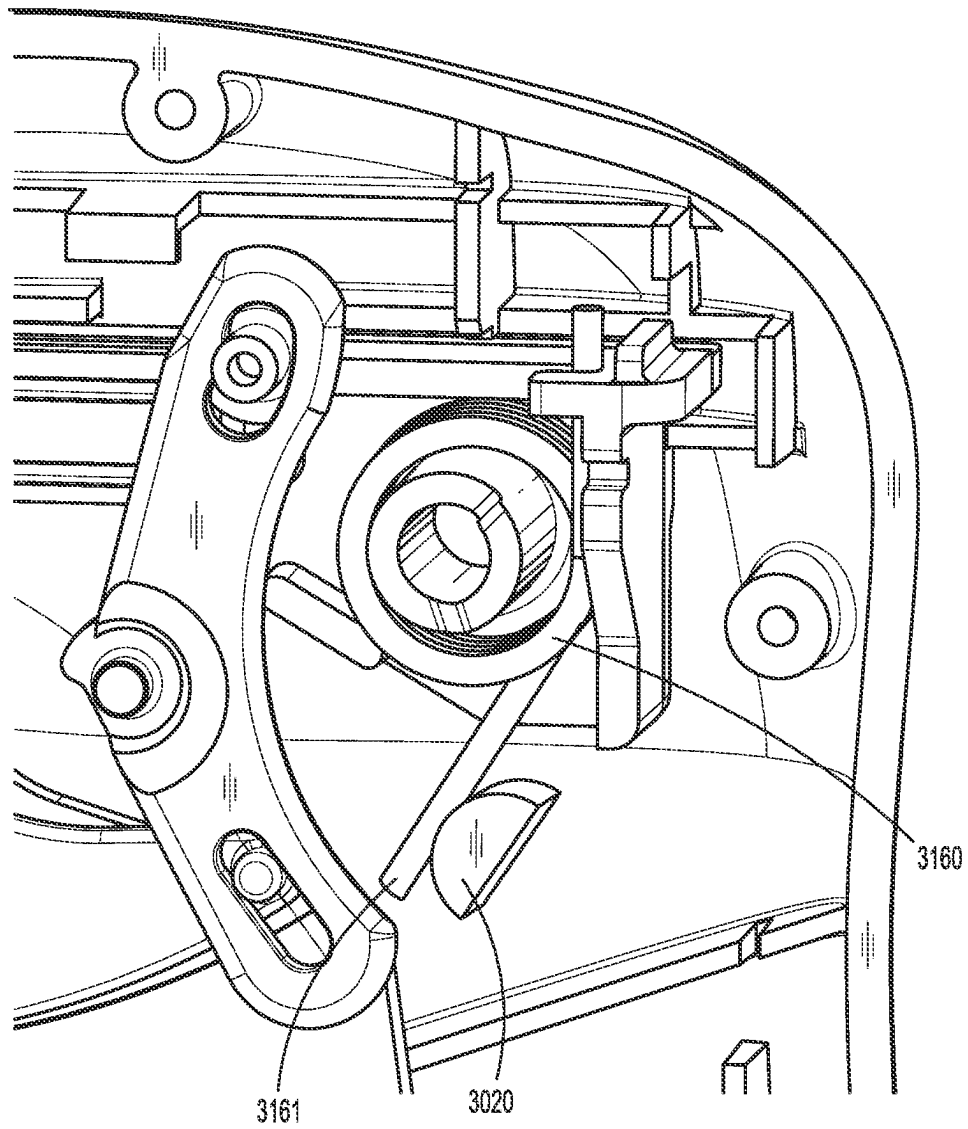


FIG. 22A

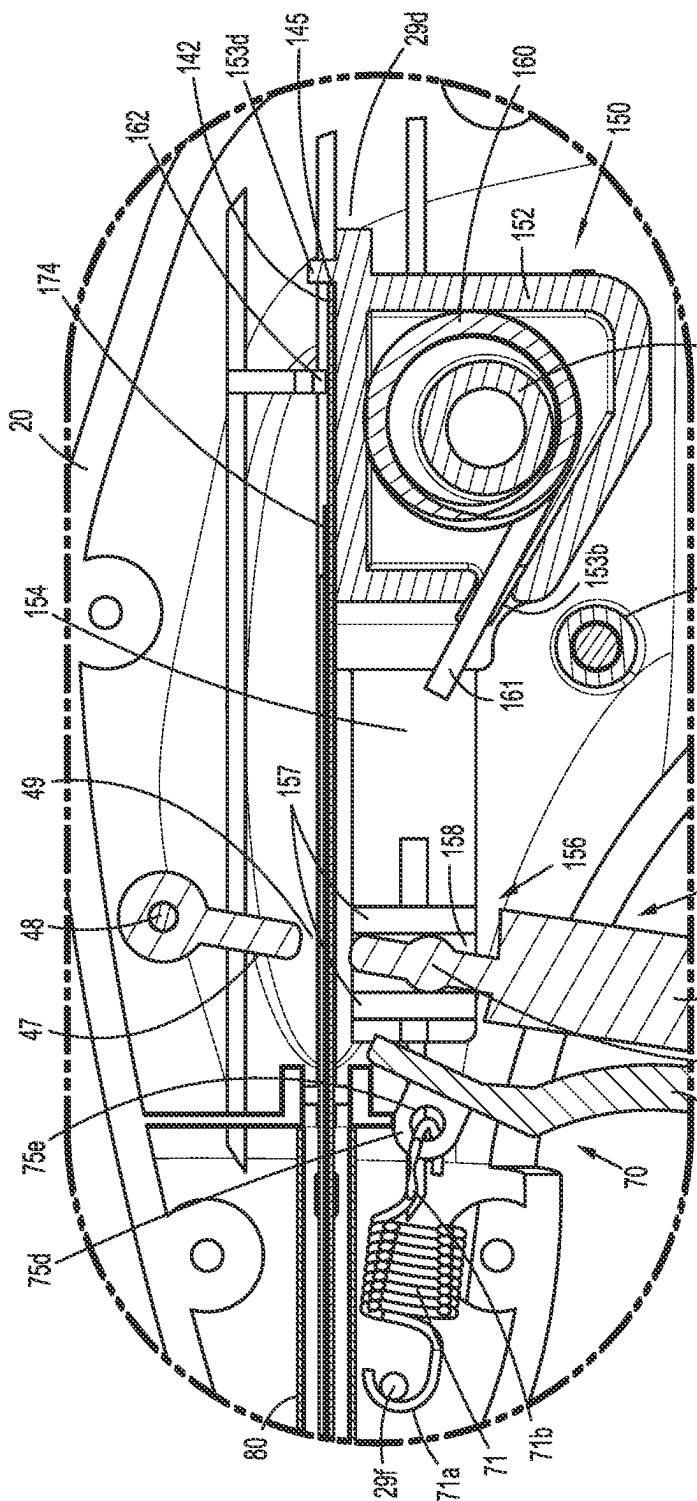


FIG. 23

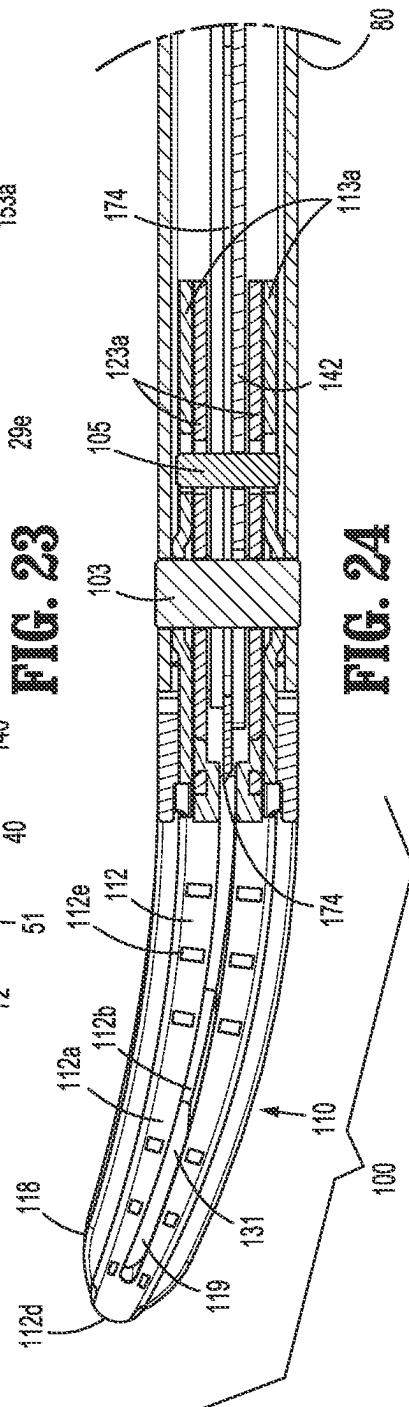


FIG. 24

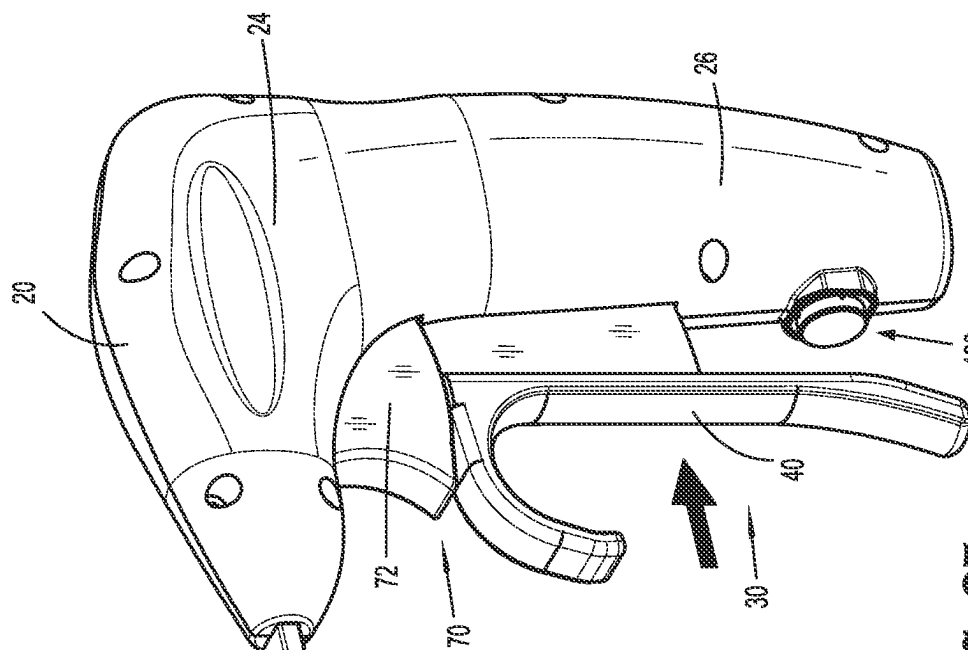


FIG. 25

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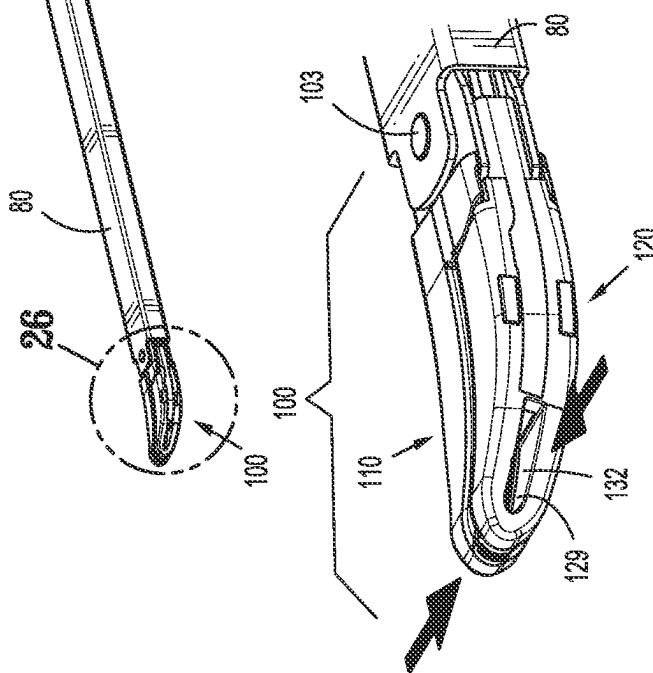


FIG. 26

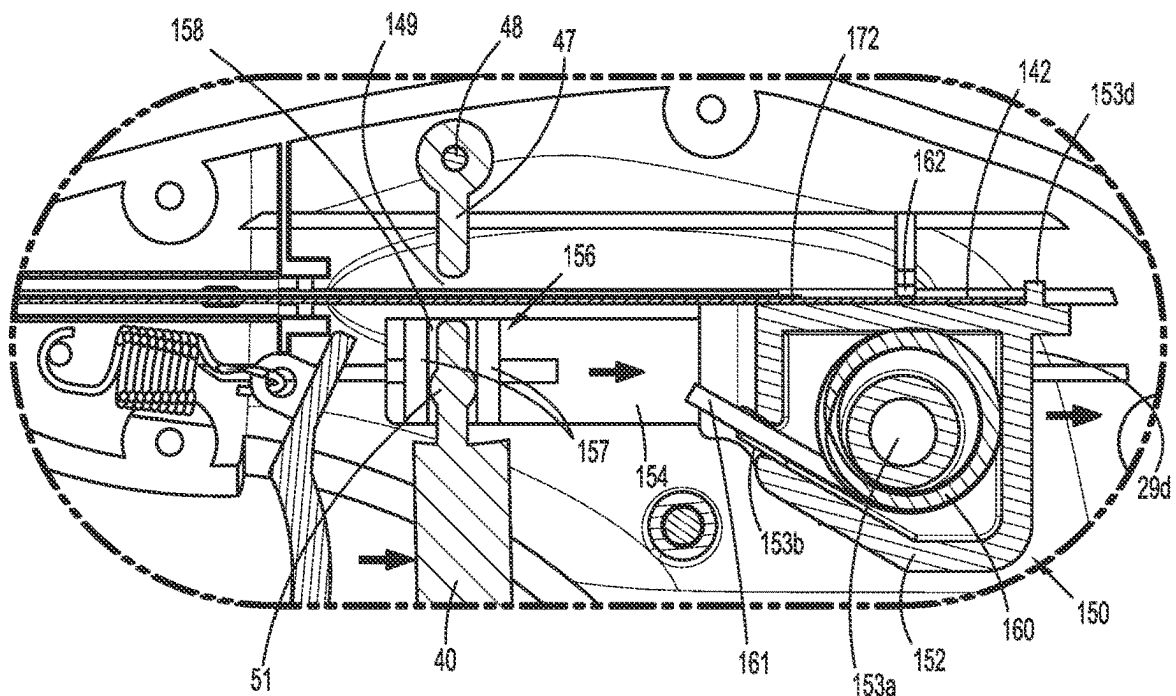
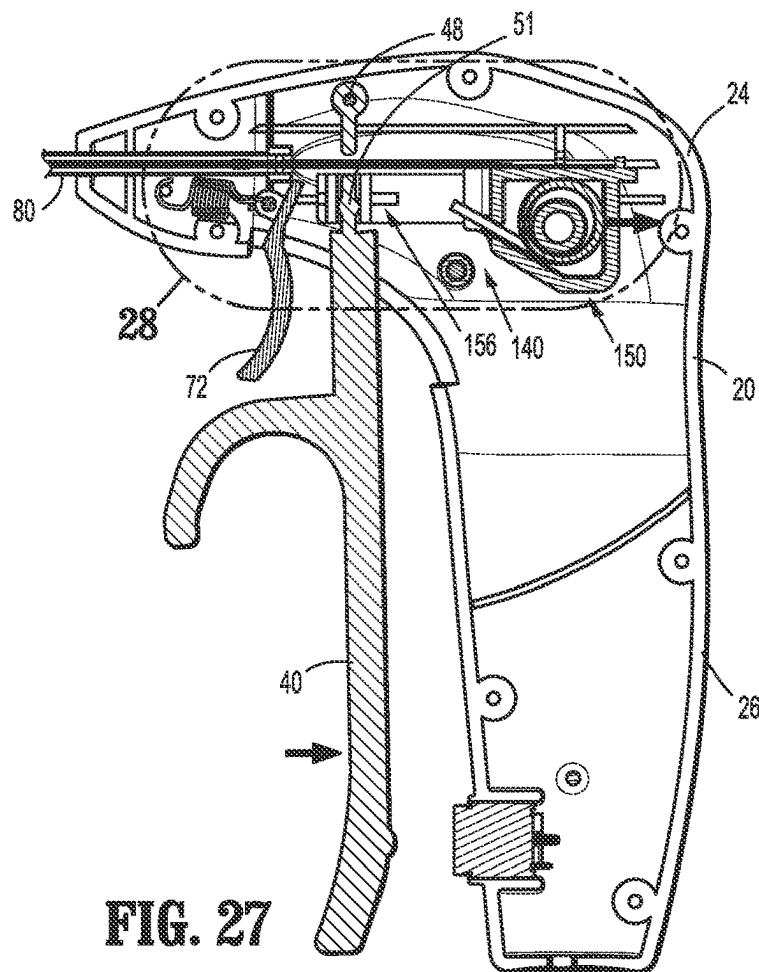


FIG. 28

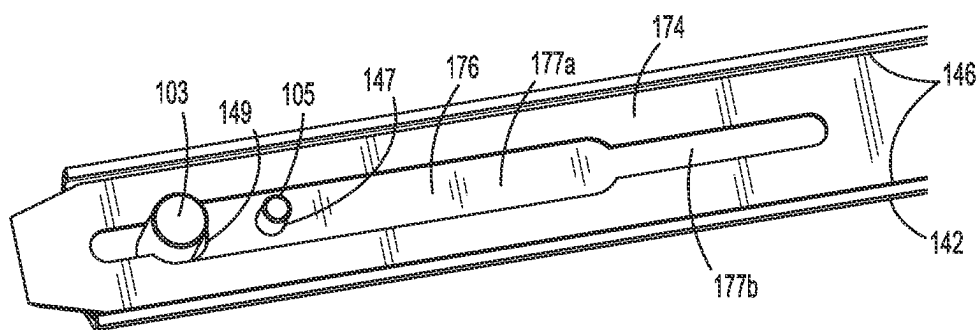


FIG. 29

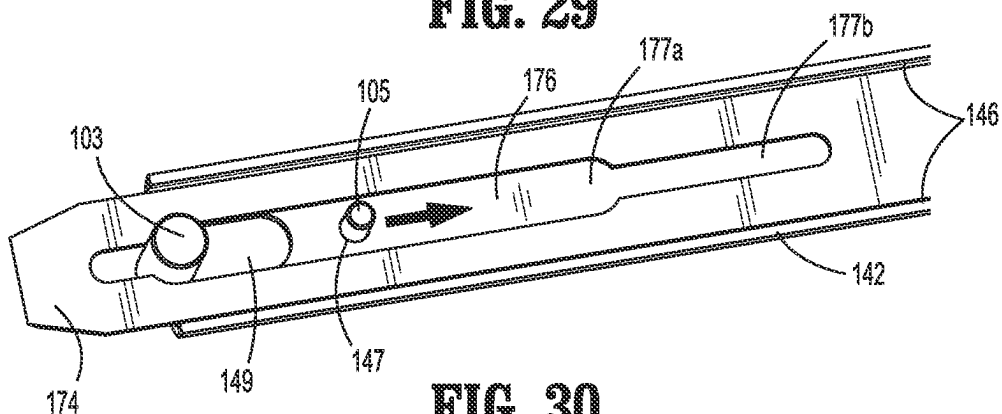


FIG. 30

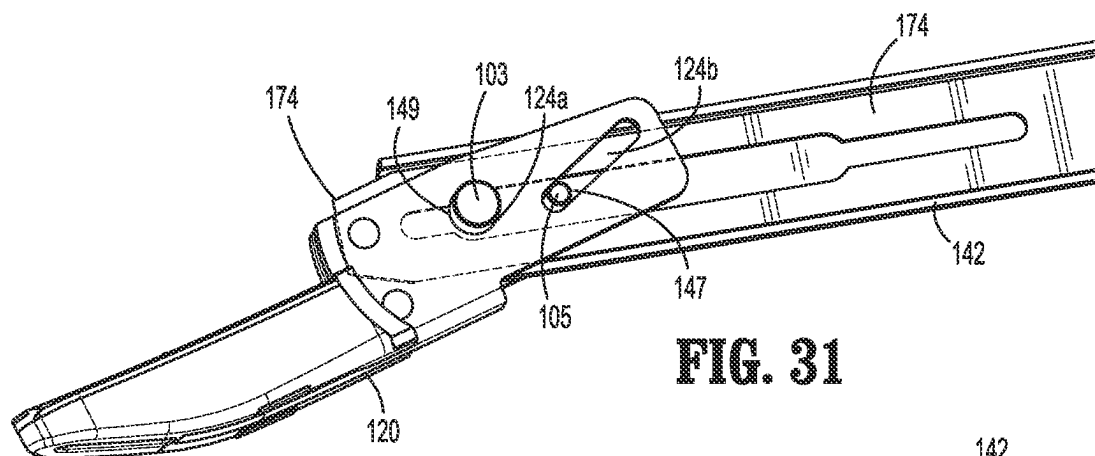


FIG. 31

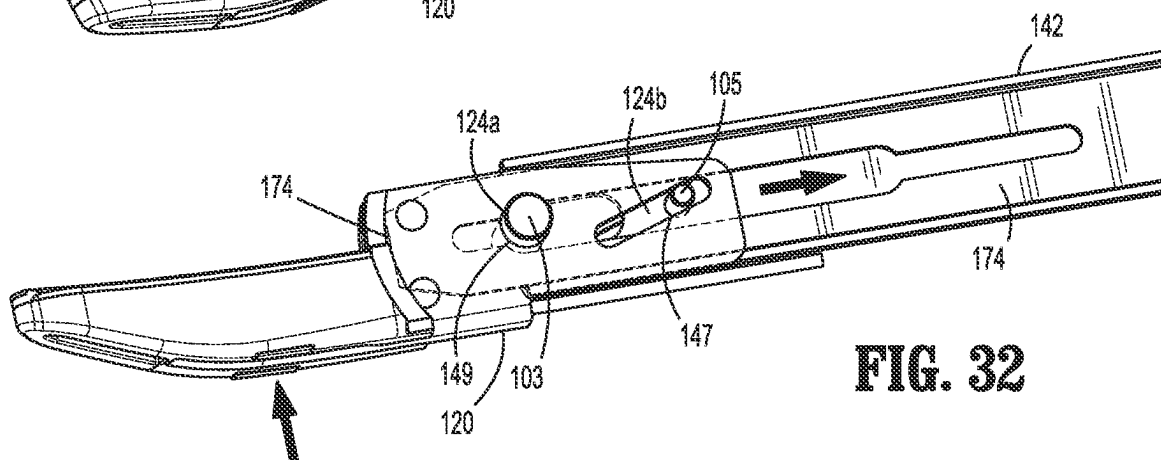
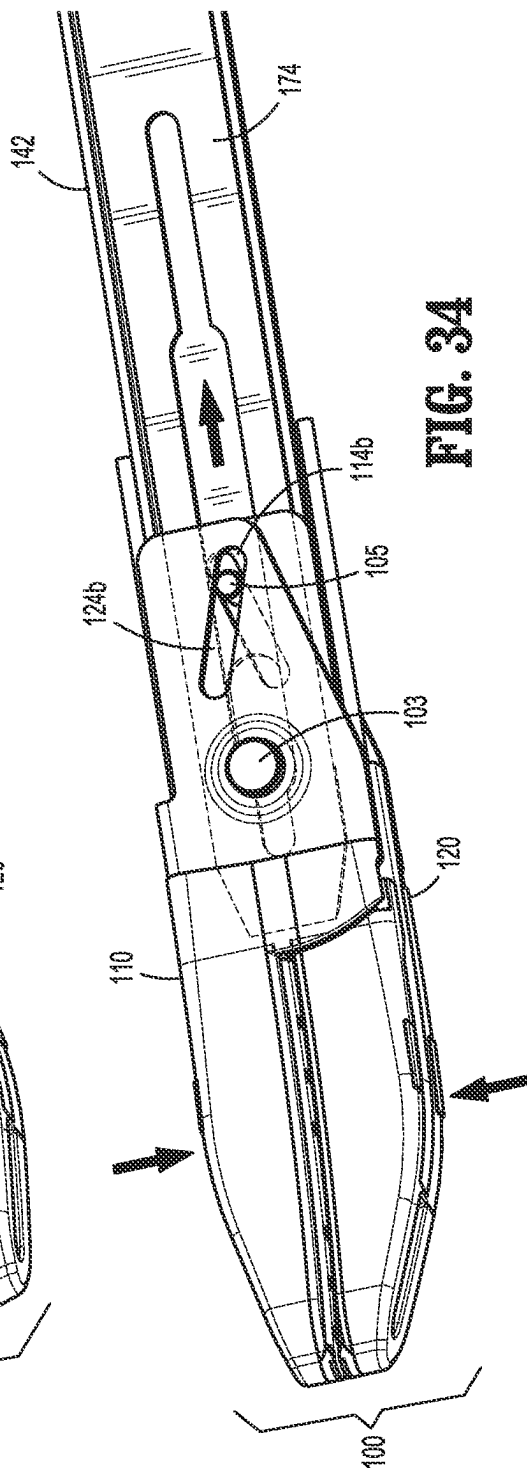
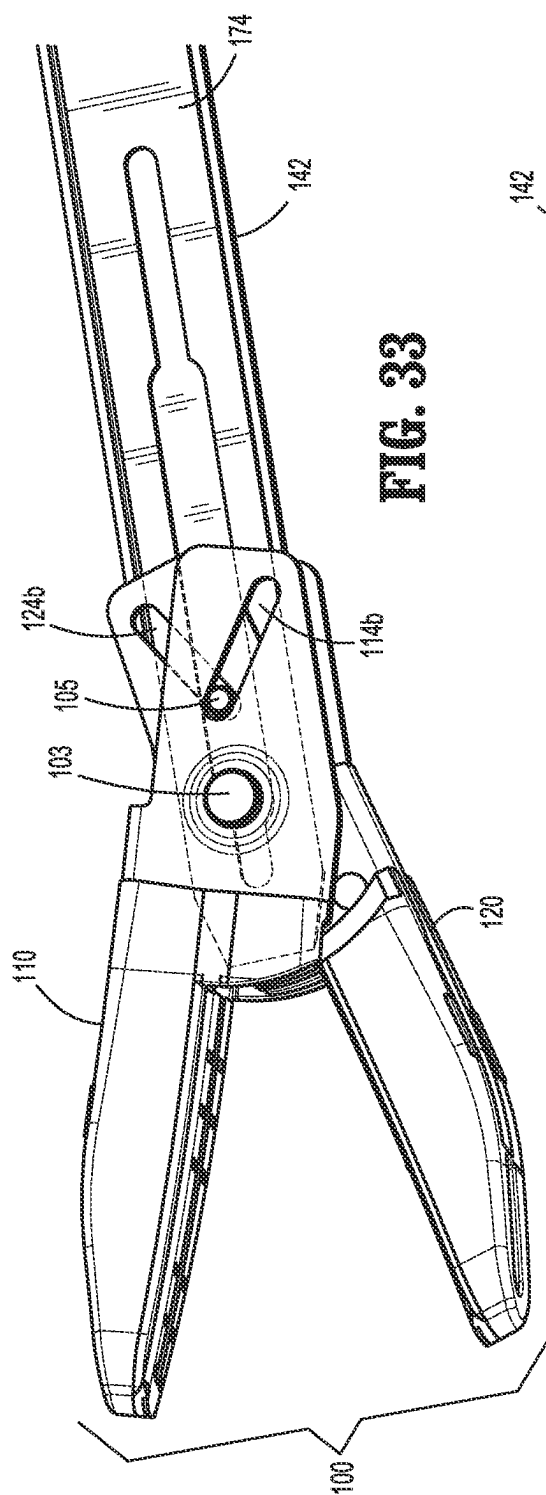


FIG. 32



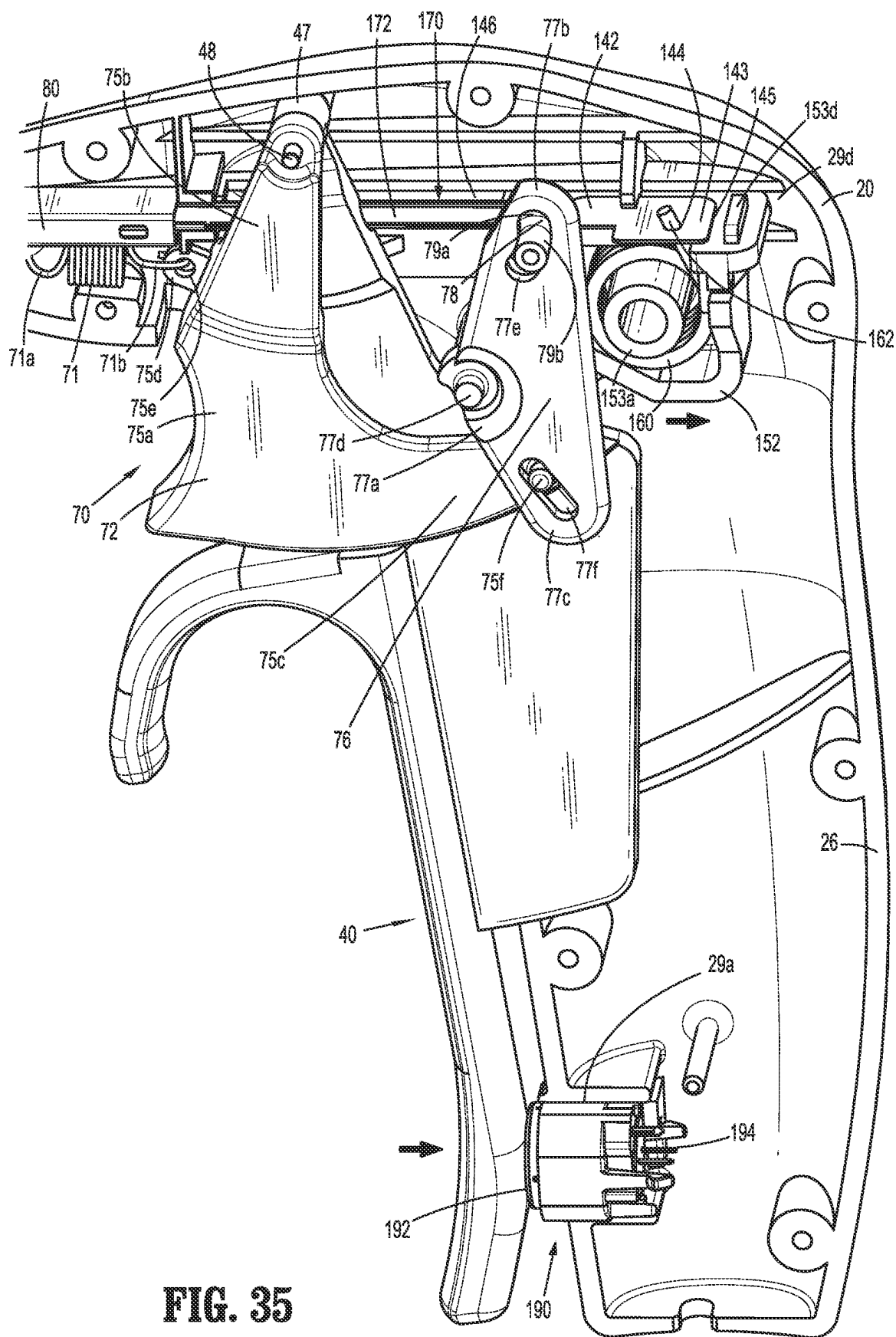


FIG. 35

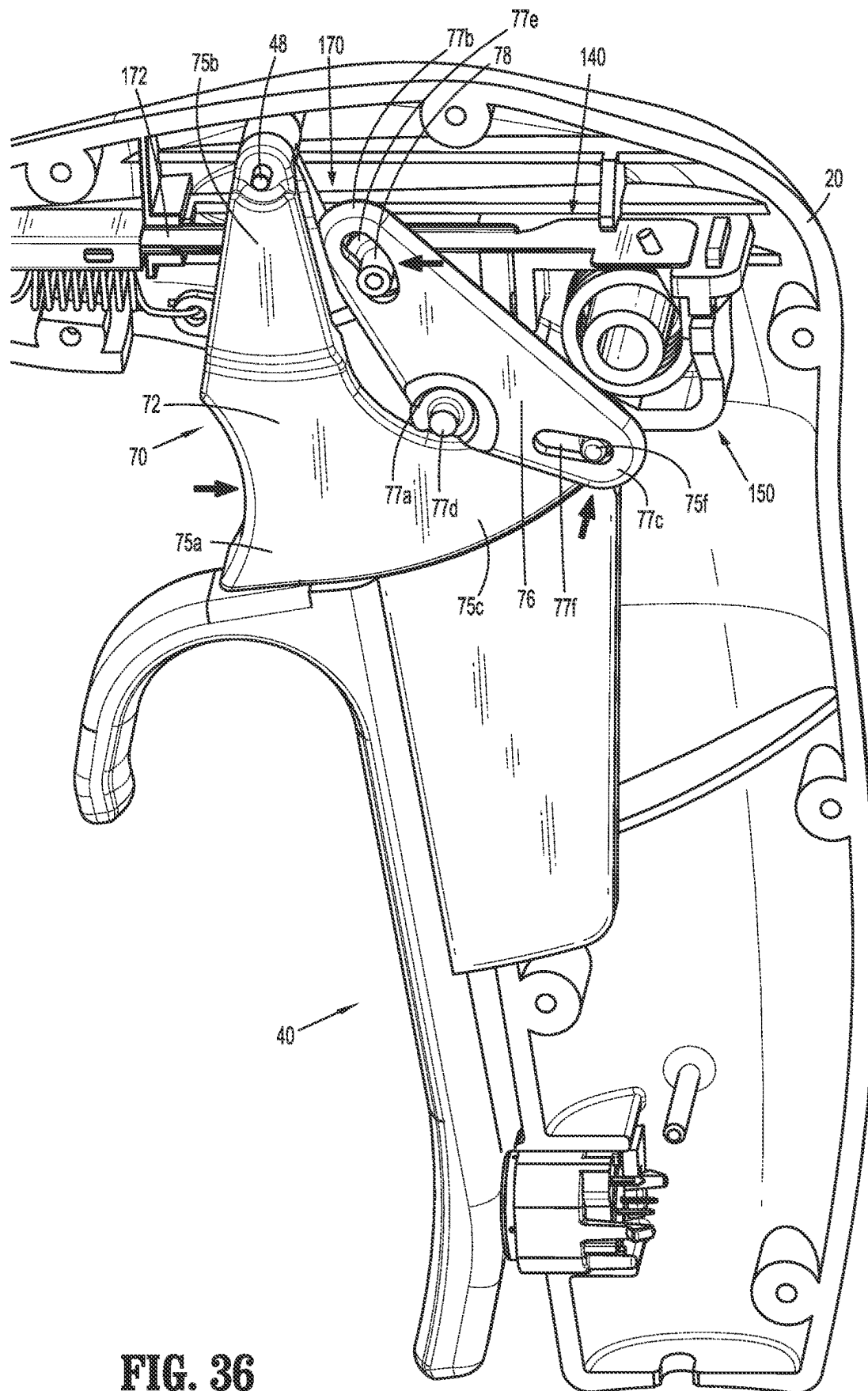


FIG. 36

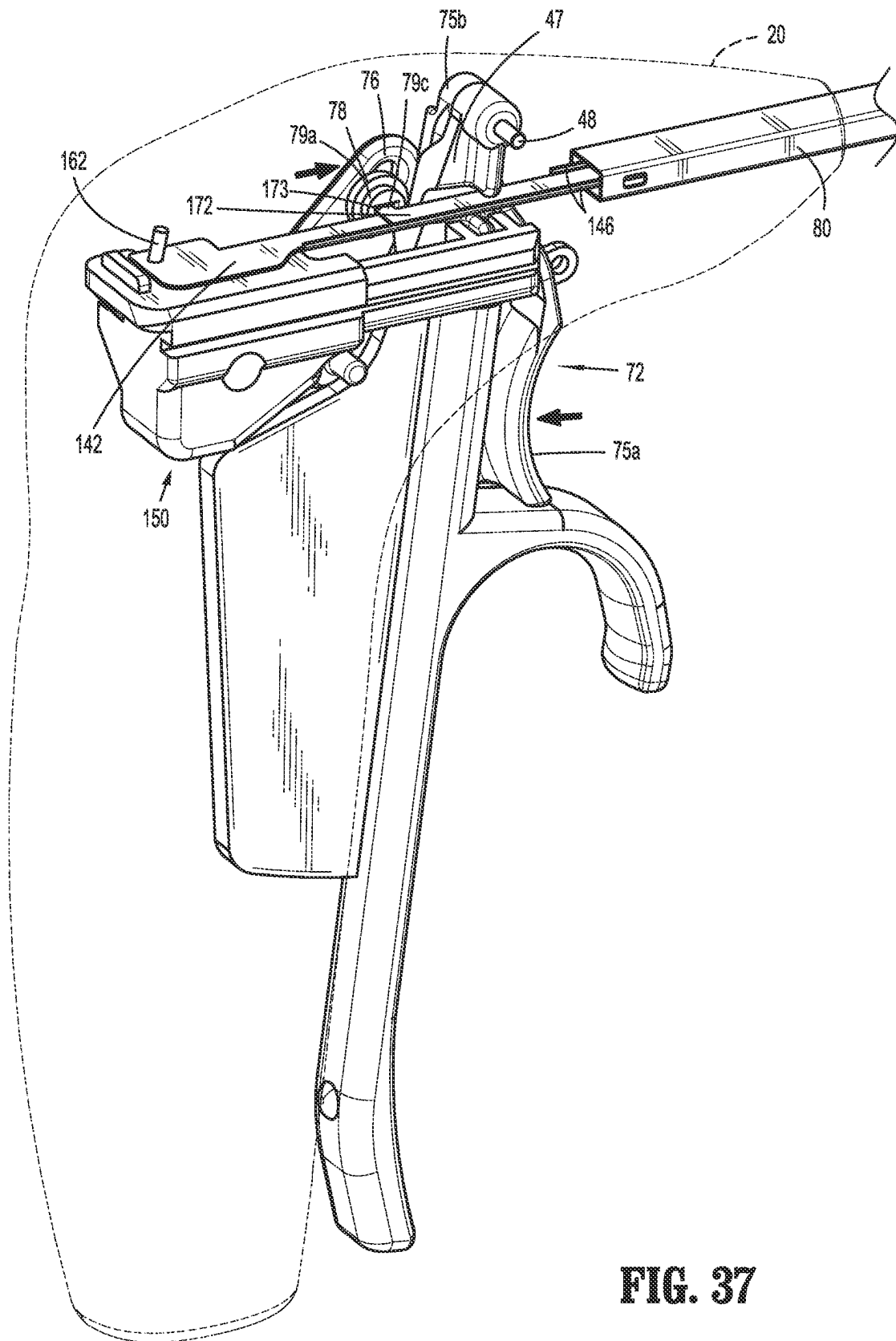
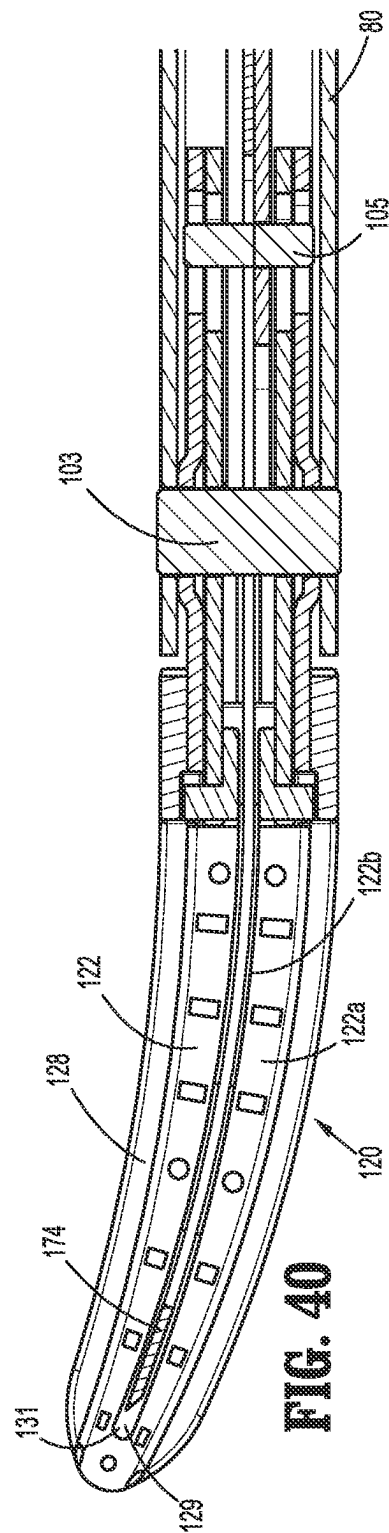
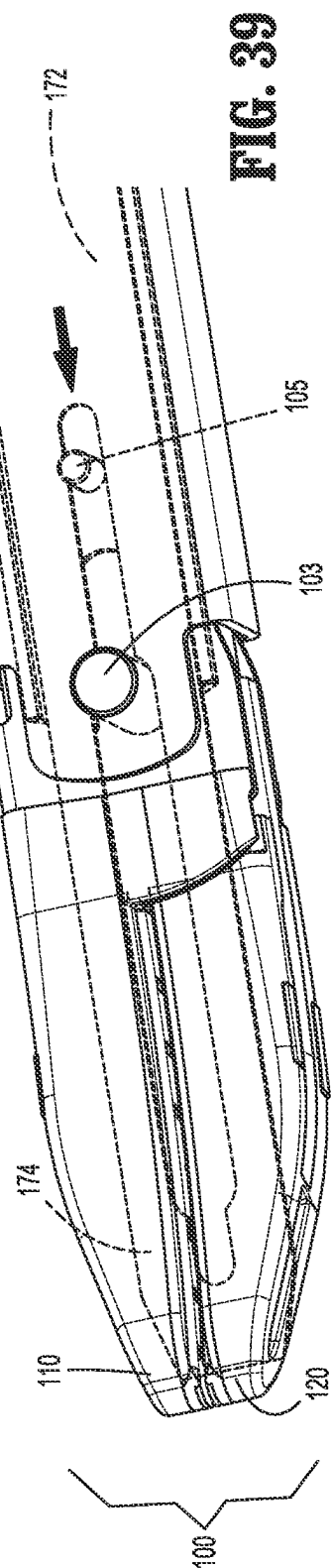
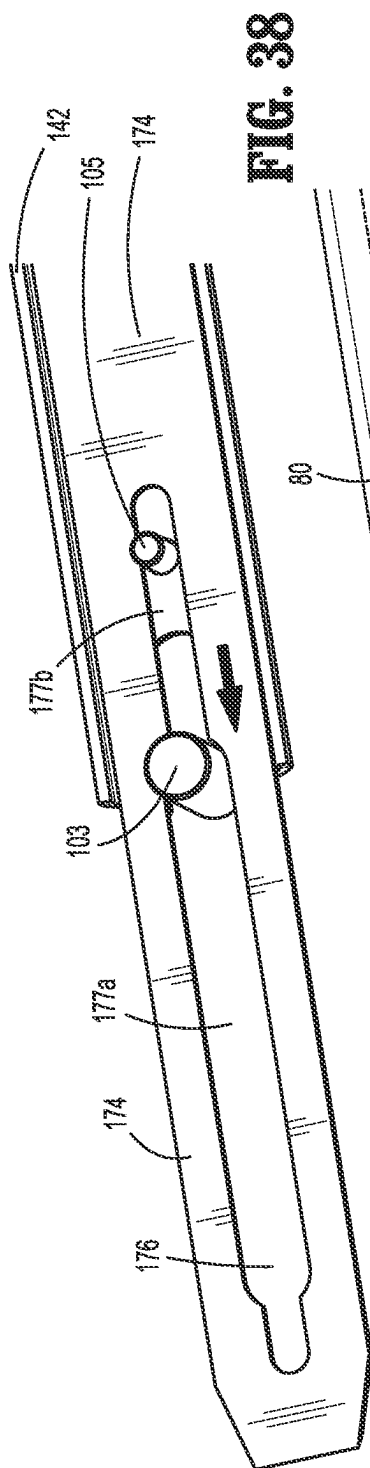


FIG. 37



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**SURGICAL INSTRUMENTS AND METHODS
FOR PERFORMING TONSILLECTOMY,
ADENOIDECTOMY, AND OTHER
SURGICAL PROCEDURES**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation application of U.S. patent application Ser. No. 17/065,033, filed on Oct. 7, 2020 and now U.S. Pat. No. 11,730,532, which is a continuation application of U.S. patent application Ser. No. 15/806,679, filed on Nov. 8, 2017 and now U.S. Pat. No. 10,828,084, which is a continuation application of U.S. patent application Ser. No. 14/719,422, filed on May 22, 2015 and now U.S. Pat. No. 9,918,779, the entire contents of each of which are hereby incorporated by reference herein.

BACKGROUND

Technical Field

The present disclosure relates to surgical instruments and methods and, more particularly, to surgical instrument and methods for performing tonsillectomy, adenoidectomy, and other surgical procedures.

Background of Related Art

The tonsils and adenoids are part of the lymphatic system and are generally located in the back of the throat. These parts of the lymphatic system are generally used for sampling bacteria and viruses entering the body and activating the immune system when warranted to produce antibodies to fight oncoming infections. More particularly, the tonsils and adenoids break down the bacteria or virus and send pieces of the bacteria or virus to the immune system to produce antibodies for fighting off infections.

Inflammation of the tonsils and adenoids (e.g., tonsillitis) impedes the ability of the tonsils and adenoids to destroy the bacteria resulting in a bacterial infection. In many instances, the bacteria remain even after treatment and serve as a reservoir for repeated infections (e.g., tonsillitis or ear infections).

A tonsillectomy and/or adenoidectomy may be performed when infections persist and antibiotic treatments fail. Persistent infection typically leads to enlarged tonsil tissue which may need to be removed since in many cases the enlarged tissue causes airway obstruction leading to various sleep disorders such as snoring or, in some cases, sleep apnea. Some individuals are also born with larger tonsils that are more prone to cause obstruction. An adenoidectomy may also be required to remove adenoid tissue when ear pain persists, or when nose breathing or function of the Eustachian tube is impaired. Often times, tonsillectomy and adenoidectomy procedures are performed at the same time.

SUMMARY

As used herein, the term “distal” refers to the portion that is being described which is further from a user, while the term “proximal” refers to the portion that is being described which is closer to a user. Further, to the extent consistent, any of the aspects described herein may be used in conjunction with any or all of the other aspects described herein.

A surgical instrument provided in accordance with the present disclosure includes a housing, a shaft extending

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distally from the housing, an end effector assembly coupled at a distal end of the shaft, a movable handle coupled to the housing, and a drive assembly operably coupling the movable handle and the end effector assembly. The drive assembly includes a drive member and a torsion spring. The drive member is configured to translate through the shaft and relative to the end effector assembly to actuate the end effector assembly. The torsion spring includes a first leg and a second leg. The first leg is operably coupled to the movable handle and configured to translate longitudinally through the housing in response to movement of the movable handle relative to the housing. The second leg is operably coupled to the drive member and configured to translate longitudinally through the housing in cooperation with the first leg to thereby transfer longitudinal movement thereof into longitudinal movement of the drive member when a force acting on the drive member is less than a threshold force. The second leg is configured to remain in fixed position, thereby tensioning the torsion spring and retaining the drive member in fixed position, in response to longitudinal movement of the first leg when the force acting on the drive member is equal to or exceeds the threshold force.

In an aspect of the present disclosure, the drive assembly further includes a slider including a housing portion operably retaining the torsion spring therein and a mandrel portion operably coupled to the movable handle. The slider is configured to translate longitudinally through the housing in response to movement of the movable handle relative to the housing.

In another aspect of the present disclosure, the first leg of the torsion spring is engaged with the housing portion of the slider and the second leg of the torsion spring is movable relative to the housing portion of the slider. In such aspects, a body portion of the torsion spring may be rotatably supported on a post disposed within the housing portion of the slider. The post is configured to enable tensioning of the torsion spring in response to movement of the first leg relative to the second leg.

In another aspect of the present disclosure, the first leg of the torsion spring is operably positioned relative to a block disposed within the housing to bias the drive member relative to the housing.

In yet another aspect of the present disclosure, the movable handle is movable relative to the housing between an initial position, a compressed position, and an activated position. The second leg is configured to translate in cooperation with the first leg in response to movement of the movable handle between the initial and compressed positions. The second leg is configured to remain in fixed position in response to movement of the movable handle between the compressed and activated positions.

In still another aspect of the present disclosure, the end effector assembly includes first and second jaw members, at least one of which is movable relative to the other between a spaced-apart position and an approximated position for grasping tissue therebetween. In such aspects, the drive member may be configured to translate through the shaft and relative to the end effector assembly to move the first and second jaw members between the spaced-apart and approximated positions.

In another aspect of the present disclosure, at least one of the first and second jaw members is adapted to connect to a source of energy for treating tissue grasped therebetween. In such aspects, an energy activation assembly disposed on the housing may also be provided. The energy activation assembly includes a switch configured to supply energy to the first and/or second jaw members. Further, the movable handle

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may be movable relative to the housing between an initial position, a compressed position, and an activated position. The second leg of the torsion spring is configured to remain in fixed position in response to movement of the movable handle between the compressed and activated positions and, in the activated position, at least a portion of the handle is configured to contact the energy activation assembly to activate the switch.

In still another aspect of the present disclosure, the second leg is configured to translate in cooperation with the first leg in response to movement of the movable handle between the initial and compressed positions for applying an appropriate closure force to tissue grasped therebetween in the compressed position. Further, the appropriate closure pressure may be maintained, e.g., via tensioning of the torsion spring, during movement of the movable handle between the compressed and activated positions.

In yet another aspect of the present disclosure, the torsion spring is pre-loaded to a less-tensioned state and wherein, in response to longitudinal movement of the first leg when the force acting on the drive member is equal to or exceeds the threshold force, the torsion spring is further tensioned to a more-tensioned state.

Another surgical instrument provided in accordance with aspects of the present disclosure includes a movable handle, a drive member, a torsion spring, and a slider. The torsion spring includes a body, a first leg, and a second leg. The second leg of the torsion spring is engaged to the drive member. The slider includes a housing and a mandrel. The housing includes a post rotatably supporting the body of the torsion spring thereon. The housing is configured to permit movement of the second leg of the torsion spring relative to the housing. The mandrel extends from the housing and is operably coupled to the movable handle such that movement of the movable handle longitudinally translates the slider. When a force acting on the drive member is less than a threshold force, the slider and the second leg of the torsion spring are configured to translate longitudinally through the housing in cooperation with one another in response to movement of the movable handle, thereby transferring longitudinal movement of the slider into longitudinal movement of the drive member. When the force acting on the drive member is equal to or exceeds the threshold force, the torsion spring is tensioned in response to movement of the movable handle such that the second leg of the torsion spring and the drive member remain in fixed position during longitudinal movement of the slider.

In an aspect of the present disclosure, the surgical instrument further includes an end effector assembly operably coupled to the drive member. In such aspects, the drive member is configured to translate relative to the end effector assembly to actuate the end effector assembly.

In another aspect of the present disclosure, the end effector assembly includes first and second jaw members. At least one of the first and second jaw members is movable relative to the other between a spaced-apart position and an approximated position for grasping tissue therebetween in response to translation of the drive member relative to the end effector assembly.

In still another aspect of the present disclosure, the force acting on the drive member corresponds to a closure pressure applied to tissue grasped between the first and second jaw members.

In yet another aspect of the present disclosure, the movable handle is movable relative to the housing between an initial position, a compressed position, and an activated position. In such aspects, the slider and the second leg of the

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torsion spring are configured to translate longitudinally through the housing in cooperation with one another in response to movement of the movable handle between the initial and compressed positions. The second leg of the torsion spring and the drive member remain in fixed position during longitudinal movement of the slider in response to movement of the movable handle between the compressed and activated positions.

In still yet another aspect of the present disclosure, at least one of the first and second jaw members is adapted to connect to a source of energy for treating tissue grasped between the first and second jaw members. In such aspects, energy may be supplied to the at least one of the first and second jaw members in response to movement of the movable handle to the activated position.

In an aspect of the present disclosure, the housing retains the first end of the torsion spring in fixed relation relative to the housing. Alternatively, the first leg of the torsion spring may be operably positioned to bias the drive member in a longitudinal direction.

BRIEF DESCRIPTION OF THE DRAWINGS

Various aspects and features of the present disclosure described herein with reference to the drawings wherein:

FIG. 1 is a perspective view of a surgical instrument provided in accordance with the present disclosure with jaw members of the end effector assembly of the surgical instrument disposed in a spaced-apart position;

FIG. 2 is a top view of the surgical instrument of FIG. 1 with the jaw members disposed in an approximated position;

FIG. 3 is a side view of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position;

FIGS. 4-6 are side views of the surgical instrument of FIG. 1 illustrating various different configurations for operably grasping the surgical instrument;

FIG. 7 is a rear, perspective view of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position and a portion of the housing removed to illustrate the internal components thereof;

FIG. 8 is a front, perspective, partially-exploded view of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position and a portion of the housing removed to illustrate the internal components thereof;

FIG. 9 is a rear, perspective view of the handle, trigger, and drive assemblies of the surgical instrument of FIG. 1 with a movable handle of the handle assembly disposed in an initial position and a trigger of the trigger assembly disposed in an un-actuated position;

FIG. 9A is a rear, perspective view of a portion of the proximal end of the surgical instrument of FIG. 1 incorporating another configuration of a drive assembly provided in accordance with the present disclosure;

FIG. 10 is a front, perspective view of the drive assembly, shaft, and end effector assembly of the surgical instrument of FIG. 1;

FIG. 11 is an enlarged, perspective view of the area of detail indicated as "11" in FIG. 10;

FIG. 11A is a perspective view of another slider assembly provided in accordance with the present disclosure and configured for use with the surgical instrument of FIG. 1;

FIG. 12 is a rear, perspective, exploded view of the drive assembly, shaft, and end effector assembly of the surgical instrument of FIG. 1;

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FIG. 13 is a transverse, cross-sectional view taken along section line "13-13" of FIG. 3;

FIG. 14 is an enlarged, perspective view of the distal end of the surgical instrument of FIG. 1;

FIG. 15 is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position adjacent tissue to be grasped;

FIG. 15A is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position pressed against tissue to be spread and/or dissected;

FIG. 15B is a side view of the distal end of the surgical instrument of FIG. 1 incorporating another configuration of jaw members provided in accordance with the present disclosure and disposed in the approximated position pressed against tissue to be spread and/or dissected;

FIG. 15B' is an enlarged, side view of the distal ends of the jaw members of FIG. 15B;

FIG. 15C is a side view of the distal end of the surgical instrument of FIG. 1 incorporating yet another configuration of jaw members provided in accordance with the present disclosure and disposed in the approximated position pressed against tissue to be spread and/or dissected;

FIG. 15C' is a top view of one of the jaw members of FIG. 15C;

FIG. 15D is a side view of the distal end of the surgical instrument of FIG. 1 incorporating still another configuration of jaw members provided in accordance with the present disclosure and disposed in the approximated position pressed against tissue to be spread and/or dissected;

FIG. 15D' is an enlarged, side view of the distal ends of the jaw members of FIG. 15D;

FIG. 16 is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the approximated position grasping tissue at the distal ends thereof;

FIG. 16A is a side view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position spreading and/or dissecting tissue;

FIG. 16B is a side view of the jaw members of FIG. 15B disposed in the spaced-apart position spreading and/or dissecting tissue;

FIG. 16C is a side view of the jaw members of FIG. 15C disposed in the spaced-apart position spreading and/or dissecting tissue;

FIG. 16D is a side view of the jaw members of FIG. 15D disposed in the spaced-apart position spreading and/or dissecting tissue;

FIG. 17 is an enlarged, perspective view of the area of detail indicated as "17" in FIG. 1;

FIG. 18 is a front, perspective view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position;

FIG. 19 is a side, perspective view of the distal end of the surgical instrument of FIG. 1 with the jaw members disposed in the spaced-apart position;

FIG. 20 is a side, perspective view of one of the jaw members of the surgical instrument of FIG. 1 with a portion thereof removed;

FIG. 21 is a transverse, cross-sectional view of the jaw member of FIG. 20;

FIG. 22 is a side, cross-sectional view taken along section line "22-22" of FIG. 1;

FIG. 22A is a side, cross-sectional view of the proximal end of the surgical instrument of FIG. 1 incorporating another configuration of the torsion spring provided in accordance with the present disclosure;

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FIG. 23 is an enlarged, side, cross-sectional view of the area of detail indicated as "23" in FIG. 22;

FIG. 24 is an enlarged, side, cross-sectional view of the area of detail indicated as "24" in FIG. 22;

FIG. 25 is a perspective view of the surgical instrument of FIG. 1 with the movable handle disposed in a compressed position and, accordingly, the jaw members disposed in the approximated position;

FIG. 26 is an enlarged, perspective view of the area of detail indicated as "26" in FIG. 25;

FIG. 27 is a side, cross-sectional view of the proximal end of the surgical instrument of FIG. 1;

FIG. 28 is an enlarged, side, cross-sectional view of the area of detail indicated as "28" in FIG. 27;

FIG. 29 is a perspective view of the distal end of the drive and knife assemblies of the surgical instrument of FIG. 1 disposed in a position corresponding to the spaced-apart position of the jaw members;

FIG. 30 is a perspective view of the distal end of the drive and knife assemblies of the surgical instrument of FIG. 1 disposed in a position corresponding to the approximated position of the jaw members;

FIG. 31 is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. 29 and further including one of the jaw members disposed in the spaced-apart position;

FIG. 32 is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. 30 and further including one of the jaw members disposed in the approximated position;

FIG. 33 is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. 29 and further including the jaw members disposed in the spaced-apart position;

FIG. 34 is a perspective view of the distal end of the drive and knife assemblies shown in FIG. 30 and further including the jaw members disposed in the approximated position;

FIG. 35 is a top, perspective view of the proximal end of the surgical instrument of FIG. 1 with the movable handle disposed in an activated position and a portion of the housing removed to illustrate the internal components thereof;

FIG. 36 is a top, perspective view of the proximal end of the surgical instrument of FIG. 1 with the trigger disposed in an actuated position and a portion of the housing removed to illustrate the internal components thereof;

FIG. 37 is a rear, perspective view of the handle, trigger, and drive assemblies of the surgical instrument of FIG. 1 with the movable handle disposed in the activated position and the trigger disposed in the actuated position;

FIG. 38 is a perspective view of the distal end of the drive and knife assemblies of the surgical instrument of FIG. 1 with the knife assembly disposed in an extended position;

FIG. 39 is a perspective view of the distal end of the drive and knife assemblies as shown in FIG. 38 and further including the jaw members disposed in the approximated position; and

FIG. 40 is a top, cross-sectional view of the distal end of the drive and knife assemblies and including the jaw members, as shown in FIG. 39.

DETAILED DESCRIPTION

Referring generally to FIGS. 1-12, a surgical instrument provided in accordance with the present disclosure is shown generally identified by reference numeral 10. Instrument 10, as described below, is configured for grasping, treating,

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and/or dissecting tissue and may find particular applicability for use in performing tonsillectomy and/or adenoidectomy procedures, although use of instrument 10 in various other surgical procedures is also contemplated and within the scope of the present disclosure.

With reference to FIGS. 1, 7, 8, and 12, instrument 10 generally includes a housing 20, a handle assembly 30, a trigger assembly 70, a shaft 80, an end effector assembly 100, a drive assembly 140, a knife assembly 170, and an energy activation assembly 190. As detailed below, shaft 80 extends distally from housing 20 and supports end effector assembly 100 at distal end 85 thereof, drive assembly 140 operably couples handle assembly 30 with end effector assembly 100 to enable selective manipulation of jaw members 110, 120 of end effector assembly 100, knife assembly 170 is operably coupled with trigger assembly 70 to enable selective translation of a knife blade 174 of knife assembly 170 relative to end effector assembly 100, and energy activation assembly 190 enables energy to be selectively delivered to end effector assembly 100.

Instrument 10 may also include an electrosurgical cable (not shown) that connects instrument 10 to a generator (not shown) or other suitable power source, although instrument 10 may alternatively be configured as a battery-powered instrument. The electrosurgical cable includes lead wires, e.g., lead wires 107 (FIG. 12), extending therethrough that have sufficient length to extend through housing 20 and shaft 80 in order to operably couple the generator, energy activation assembly 190, and end effector assembly 100 with one another to enable the selective supply of energy to electrically-conductive plates 112, 122 of jaw members 110, 120 of end effector assembly 100, e.g., upon activation of activation switch 194 of energy activation assembly 190.

Referring to FIGS. 1-7, housing 20 houses the internal working components of instrument 10 and is formed from first and second housing components 22a, 22b configured to engage one another via a plurality of pin-aperture engagements 23 spaced around the perimeter of housing 20. Housing 20 defines a pistol-style configuration having a longitudinally-extending barrel portion 24 and a fixed handle portion 26 that extends from barrel portion 24 in generally perpendicular orientation relative thereto.

Housing 20, movable handle 40 of handle assembly 30, and trigger 72 of trigger assembly 70 are ergonomically configured to enable operable grasping of instrument 10 in a plurality of different positions. Housing 20, more specifically, defines an elongated indentation 27 within barrel portion 24 on either side thereof, while a waist 28 is recessed annularly about handle portion 26 adjacent the interconnection between handle portion 26 and barrel portion 24. Movable handle 40, more specifically, defines a grasping portion 42 having an elongated proximal leg 43 that extends the length of fixed handle portion 26 of housing 20, a proximal foot 44 disposed at the free end of proximal leg 43 and angled distally relative to proximal leg 43, and an arcuate segment 45 disposed at the opposite end of proximal leg 43 and extending distally therefrom. Arcuate segment 45 culminates in a distal tail 46 and defines a sufficient diameter so as to operably receive a user's finger between distal tail 46 and proximal leg 43. Trigger 72, more specifically, includes a concave trigger surface 73 defining a saddle 74 configured to help retain a user's finger therein.

With particular reference to FIGS. 4-6, and initially to FIG. 4, in a first operable grasping position, the user's hand is positioned such that the thumb is partially received within waist 28, the tip of the index finger extends across trigger 72 and is partially received within saddle 74, the middle finger

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extends across movable handle 40 and is positioned adjacent arcuate segment 45 of movable handle 40 between proximal leg 43 and distal tail 46, and the ring finger and pinky are positioned distally of and adjacent to proximal leg 43. In this position, waist 28 inhibits slipping of the thumb, saddle 74 inhibits slipping of the index finger, proximal leg 43 and distal tail 46 retain the middle finger therebetween to enable both proximal and distal movement of movable handle 40, proximal leg 43 provides a surface against which the ring finger and pinky can be utilized to urge movable handle 40 proximally, and proximal foot 44 inhibits slipping of the ring finger and pinky off the free end of movable handle 40.

Referring to FIG. 5, in a second operable grasping position, the user's hand is positioned such that the index finger is partially received within elongated indentation 27 on the opposite side of housing 20, the middle finger extends transversely across trigger 72 and is partially received within saddle 74, the ring finger extends across movable handle 40 and is positioned adjacent arcuate segment 45 of movable handle 40 between proximal leg 43 and distal tail 46, and the pinky is positioned distally of and adjacent to proximal leg 43. In this position, elongated indentation 27 inhibits slipping of the index finger, saddle 74 inhibits slipping of the middle finger, proximal leg 43 and distal tail 46 retain the ring finger therebetween to enable both proximal and distal movement of movable handle 40, and proximal leg 43 provides a surface against which the pinky can be utilized to urge movable handle 40 proximally.

Referring to FIG. 6, in a third operable grasping position, the user's hand is positioned such that the thumb is wrapped around a free end of fixed handle portion 26 of housing 20, the index finger and middle finger are positioned proximally of and adjacent the free end of proximal leg 43 of movable handle 40, the ring finger extends across movable handle 40 and is positioned adjacent arcuate segment 45 of movable handle 40 between proximal leg 43 and distal tail 46, and the pinky extends across trigger 72 and is partially received within saddle 74. In this position, proximal foot 44 inhibits slipping of the index finger off the free end of movable handle 40, proximal leg 43 and distal tail 46 retain the ring finger therebetween to enable both proximal and distal movement of movable handle 40, proximal leg 43 provides a surface against which the index and middle fingers can be utilized to urge movable handle 40 proximally, and saddle 74 inhibits slipping of the pinky.

With reference to FIG. 7, fixed handle portion 26 of housing 20 defines a bay 29a configured to receive and support energy activation assembly 190, which is operable to initiate and terminate the delivery of energy to end effector assembly 100. Energy activation assembly 190 includes a depressible button 192 that is mechanically coupled to a switch 194 mounted within bay 29a of fixed handle portion 26 and is engagable by a button activation post 196 extending proximally from a proximal side of movable handle 40 upon movement of the movable handle 40 to the activated position, as detailed below. Switch 194 is configured to electrically communicate with end effector assembly 100 and the generator (not shown) via suitable electrical wiring, e.g., leads 107 (FIG. 12), extending through housing 20, shaft 80, and/or an external cable (not shown) to enable energy to be supplied from the generator (not shown) to end effector assembly 100 upon activation of switch 194.

Referring additionally to FIG. 8, barrel portion 24 of housing 20 defines a distal aperture 29b (FIG. 7) configured to receive proximal end 82 of shaft 80 therein, and an engagement feature (not shown) extending inwardly from

each of first and second housing components **22a**, **22b** for receipt within opposed apertures **83** defined through proximal end **82** of shaft **80** for securing proximal end **82** of shaft **80** within barrel portion **24** of housing **20**. Shaft **80** extends distally from housing **20** and defines a generally rectangular cross-sectional configuration oriented such that the larger width dimension thereof extends laterally and the smaller height dimension thereof extends vertically. This configuration of shaft **80** relative to the orientation of jaw members **110**, **120** (FIG. 7) provides an enhanced “line-of-sight” for visualizing the surgical site adjacent end effector assembly **100**. As described in greater detail below, shaft **80** includes a pair of spaced-apart clevis members **84** extending from the top and bottom walls, e.g., the larger width dimension walls, of shaft **80** at distal end **85** thereof, each of which defines an aperture **86** for receiving a pivot pin **103** to operably support end effector assembly **100** at distal end **85** of shaft **80**. In this configuration, apertures **86** are vertically-aligned with one another. Shaft **80** further includes, as noted above, opposed apertures **83** defined through the side walls, e.g., the smaller height dimension walls, of shaft **80** at proximal end **82** thereof for receiving engagement features (not shown) extending inwardly from first and second housing components **22a**, **22b** to secure proximal end **82** of shaft **80** within housing **20**.

Barrel portion **24** of housing further includes a pair of pivot apertures **29c**, a longitudinal track **29d** (FIG. 13), a pivot boss **29e**, and a retention pin **29f**. Each pivot aperture **29c** is defined on the inwardly-facing surface of one of first and second housing components **22a**, **22b** (only pivot aperture **29c** of first housing component **22a** is shown) and is configured to receive pivot pin **48** to pivotably couple movable handle **40** and trigger **72** to housing **20**. Longitudinal track **29d** (FIG. 13) is defined on the inwardly-facing surface of first housing component **22a** and is configured to guide translation of drive assembly **140** relative to housing **20**. Pivot boss **29e** extends inwardly from first housing component **22a** and is configured to pivotably couple linkage **76** of trigger assembly **70** to housing **20**. Retention pin **29f** extends inwardly from first housing component **22a** and is configured to retain a fixed end **71a** of biasing member **71** of trigger assembly **70** in fixed position relative to housing **20**. The importance of these features of barrel portion **24** of housing **20** will become more apparent in view of the description below.

Turning to FIGS. 7-13, handle assembly **30** includes a movable handle **40** that is movable relative to fixed handle portion **26** of housing **20** between an initial position, a compressed position, and an activated position, as explained in greater detail below, to impart movement of jaw members **110**, **120** of end effector assembly **100** between a spaced-apart position and an approximated position for grasping tissue therebetween and for initiating the supply of energy to end effector assembly **100** for treating grasped tissue. Movable handle **40** includes grasping portion **42**, detailed above, which extends from housing **20** adjacent fixed handle portion **26**, and flange portion **47**, which extends upwardly into housing **20**. Flange portion **47** is pivotably coupled within housing **20** at the free end of flange portion **47** via pivot pin **48**. Pivot pin **48** is engaged within and extends between pivot apertures **29c** of first and second housing components **22a**, **22b** of housing **20** to permit movable handle **40** to pivot about pivot pin **48** and relative to housing **20** between the initial position, the compressed position, and the activated position. Pivot pin **48** is disposed on one side of, e.g., above, drive assembly **140**, while grasping portion **42** of movable handle **40** is disposed on the other side of, e.g., below, drive

assembly **140**, to provide a mechanical advantage when actuating movable handle **40**.

Flange portion **47** of movable handle **40** further includes a cut-out **49** defined therein and an engagement bulge **51** protruding therefrom. Cut-out **49** is configured to slidably receive drive plate **142** of drive assembly **140** and knife plate **172** of knife assembly **170**. Engagement bulge **51** is configured to operably engage flange portion **47** of movable handle **40** with slider assembly **150** of drive assembly **140**, as detailed below.

Drive assembly **140** includes drive plate **142** and slider assembly **150**. Drive plate **142** extends distally from housing **20** and through shaft **80** to operably engage end effector assembly **100** such that, as detailed below, translation of drive plate **142** through shaft **80** and relative to end effector assembly **100** pivots jaw members **110**, **120** of end effector assembly **100** between the spaced-apart and approximated positions. Slider assembly **150** operably couples flange portion **47** of movable handle **40** with drive plate **142** such that pivoting of movable handle **40** between the initial position and the compressed position pivots jaw members **110**, **120** of end effector assembly **100** between the spaced-apart and approximated positions, while ensuring application of an appropriate closure force or closure force within an appropriate closure force range to tissue grasped between jaw members **110**, **120** in the approximated position thereof.

Slider assembly **150** includes a proximal housing **152**, a distal extension **154** extending distally from proximal housing **152**, and a mandrel **156** disposed at the distal end of distal extension **154**. Proximal housing **152** includes a post **153a** configured to receive a torsion spring **160** thereabout, a first slot **153b** configured to retain a first leg **161** of torsion spring **160** therein in fixed relation relative thereto, and a second slot **153c** configured to operably receive second leg **162** of torsion spring **160** therein. Proximal housing **152** further includes an abutment rib **153d** disposed thereon adjacent second slot **153c**, and a flange member **153e** configured for receipt within longitudinal track **29d** (FIG. 13) of first housing component **22a** of housing **20**.

Referring briefly to FIG. 11A, another slider assembly **2150** provided in accordance with the present disclosure is similar to slider assembly **150** (FIGS. 10-11) except for the configuration of the proximal housing and spring; thus, only these differences are detailed below. Proximal housing **2152** of slider assembly **2150** is configured to house a compression spring **2160** therein. Compression spring **2160** defines a first end **2161** and a second end **2162**. First end **2161** of compression spring **2160** is engaged with a vertical plate. Second end **2162** of compression spring **2160** is engaged with an inner wall of proximal housing **2152**. In use, compression spring **2160** functions similar to torsion spring **160** (FIGS. 10-11), as detailed below, except that, rather than being further tensioned via application of a torsional force thereto, compression spring **2160** is further tensioned via application of a compressive force thereto.

Returning to FIGS. 7-13, mandrel **156**, as noted above, is disposed at the distal end of distal extension **154** of slider assembly **150**. Mandrel **156** includes a pair of spaced-apart walls **157** defining a channel **158** therebetween. Channel **158** is configured to receive engagement bulge **51** of flange portion **47** of movable handle **40** while permitting vertical sliding of engagement bulge **51** within channel **158**. As a result of this configuration, upon pivoting of movable handle **40** between the initial, compressed, and activated positions, engagement bulge **51** is urged into contact with one of the walls **157** defining channel **158** to thereby translate slider assembly **150** within housing **20**. The vertical sliding of

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engagement bulge **51** within channel **158** during such urging ensures that slider assembly **150** is translated longitudinally within and relative to housing **20** despite the arcuate travel of movable handle **40** as movable handle is pivoted about pivot pin **48** relative to housing **20**.

Drive plate **142** includes a flange **143** disposed at the proximal end thereof. Flange **143** defines an aperture **144** configured to receive second leg **162** of torsion spring **160** therein such that translation of second leg **162** of torsion spring **160** relative to housing **20** effects corresponding translation of drive plate **142** relative to housing **20**. With respect to slider assembly **2150**, vertical plate **2163** is engaged within a slot **2144** defined within drive plate **142** (see FIG. **11A**), and functions in a similar manner as detailed below with respect to slider assembly **150**. Flange **143** further defines a proximal edge **145** configured to abut abutment rib **153d** of proximal housing **152** in a proximal-most position of drive plate **142** relative to slider assembly **150** to inhibit further proximal movement of drive plate **142** relative to slider assembly **150**.

Drive plate **142**, as mentioned above, extends distally from housing **20** and through shaft **80** to operably engage end effector assembly **100**. Drive plate **142** is oriented similarly to shaft **80**, e.g., such that the width of drive plate **142** extends along the width dimension of shaft **80**. Drive plate **142** further defines a track edge **146** extending along a portion of each longitudinal side thereof. Track edges **146** are configured to slidably receive knife plate **172**, as detailed below. A cam-pin aperture **147** configured to receive a cam pin **105** associated with end effector assembly **100** is defined transversely through drive plate **142** towards the distal end of drive plate **142**. A mouth **149** configured to receive a pivot pin **103** associated with end effector assembly **100** is defined at the distal end of drive plate **142**.

With momentary reference to FIG. **9A**, as an alternative to providing a slider assembly that operably retains a torsion spring therein for coupling to drive plate **142**, a torsion spring **1160** may be operably coupled between movable handle **40** and drive plate **142** without a slider assembly. More specifically, in some embodiments, torsion spring **1160** is mounted about a post **1153a** extending transversely from movable handle **40** and includes a first leg **1161** and a second leg **1162**. First leg **1161** of torsion spring **1160** is configured for receipt within an aperture **1144** defined through drive plate **142** to operably couple movable handle **40** with drive plate **142**, while second leg **1162** of torsion spring **1160** is fixed relative to movable handle **40** via abutment with a protrusion **1153c** thereof. In use, torsion spring **1160** operates similarly to torsion spring **160** (FIGS. **10-11**) and, thus, a separate description of the use of torsion spring **1160** is omitted as being superfluous.

With reference to FIGS. **14-21**, as mentioned above, end effector assembly **100** is operably supported at distal end **85** of shaft **80** and includes opposing jaw members **110**, **120** pivotably coupled to one another and movable relative to one another and shaft **80** between a spaced-apart position and an approximated position for grasping tissue therebetween. Each jaw member **110**, **120** includes an electrically-conductive plate **112**, **122**, a jaw frame **113**, **123**, a spacer **115** (only spacer **115** of jaw member **110** is shown (FIG. **21**)), and an outer housing **118**, **128**, each of which is detailed below. Jaw members **110**, **120** define curved configurations, wherein jaw members **110**, **120** bend upwardly from a longitudinal axis of shaft **80**, e.g., towards the upper, larger width dimension wall of shaft **80**. This configuration facilitates use of instrument **10** in tonsillectomy and adenoidectomy procedures as well as other surgical procedures and

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allows for increased visualization of the surgical site in these and other procedures. Except where specifically noted otherwise, jaw members **110**, **120** define mirror-image configurations of one another.

Jaw frames **113**, **123** of jaw members **110**, **120** each include a pair of spaced-apart proximal flanges **113a**, **123a** and a distal jaw support **113b**, **123b**. Jaw frames **113**, **123** are formed via stamping and made from stainless steel, although other manufacturing processes and/or materials for forming jaw frames **113**, **123** are also contemplated. Proximal flanges **113a** of jaw member **110** are spaced-apart further than proximal flanges **123a** of jaw member **120** so as to allow proximal flanges **123a** of jaw member **120** to be positioned between proximal flanges **113a** of jaw member **110** during assembly. Further, the proximal flanges **113a**, **123a** of each pair define aligned pivot apertures **114a**, **124a** and aligned cam slots **114b**, **124b**. Pivot pin **103** of end effector assembly **100** is configured for vertical insertion through apertures **86** of clevis members **84** of shaft **80** and pivot apertures **114a**, **124a** to pivotably couple jaw members **110**, **120** to shaft **80** and one another with jaw members **110**, **120** being laterally movable, e.g., along the larger width dimension of shaft **80**, between the spaced-apart and approximated positions. Pivot pin **103** is configured to at least partially enter mouth **149** of drive plate **142** to permit drive plate **142** to slide further distally relative to end effector assembly **100** to a position wherein mouth **149** of drive plate **142** at least partially surrounds pivot pin **103**.

The cam slots **114b** of proximal flanges **113a** of jaw member **110** are oppositely angled relative to the cam slots **124b** of proximal flanges **123a** of jaw member **120**. Cam pin **105** of end effector assembly **100** is configured for insertion through each cam slot **114b**, **124b** as well as cam-pin aperture **147** of drive plate **142** to operable couple drive plate **142** with jaw members **110**, **120** such that translation of drive plate **142** relative to jaw members **110**, **120** pivots jaw members **110**, **120** about pivot pin **103** and relative to one another and shaft **80** between the spaced-apart and approximated positions.

With particular reference to FIGS. **19-21**, although only the features of jaw member **110** or jaw member **120** are described below and/or illustrated in the figures, it is noted that jaw members **110**, **120** defines a mirror-image configurations of one another (unless specifically contradicted herein) and, thus, any description and/or illustration of one jaw member **110**, **120** applies similarly to the other jaw member **110**, **120**.

Distal jaw support **113b** of jaw frame **113** of jaw member **110** extends distally from proximal flange **113a** and defines a generally “L-shaped” configuration. Distal jaw support **113b** is configured to support electrically-conductive plate **112**, spacer **115**, and outer housing **118** of jaw member **110** thereon. However, distal jaw support **113b** do not extend distally the entire length of jaw member **110**. Rather, distal jaw support **113b** defines a length of about 50% to about 75% of the lengths of electrically-conductive plate **112**, spacer **115**, and outer housing **118** such that about 25% to about 50% of the lengths of these components extend distally beyond distal jaw support **113b**.

Spacer **115** of jaw member **110** defines a generally “M-shaped” configuration, is formed from an electrically-insulative material, and is overmolded onto distal jaw support **113b** during a first overmold, although other manufacturing processes are also contemplated. Spacer **115** defines a body **116a** and a pair of wings **116b** surrounding body **116a**. Spacer **115** is positioned to electrically-isolate electrically-conductive plate **112** and distal jaw support **113b**

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from one another. A knife slot **116c** extends longitudinally through body **116a** of spacer **115** and is generally centered relative to body **116a**. Knife slot **116c** is open only to the top of spacer **115**, except for the distal portion thereof, which extends beyond distal jaw support **113b** and is open on both the top and bottom sides thereof to provide a window **116d**. A support-receiving channel **116e** extends longitudinally through body **116a** at a position laterally offset relative to knife slot **116c** so as to not interfere therewith. Support-receiving channel **116e** is open to the bottom of spacer **115** and is configured to receive the upright of the “L-shaped” distal jaw support **113b** upon the first overmolding of spacer **115** thereabout. Body **116a** of spacer **115** further defines a tunnel **116f** configured to permit passage of lead wire **107** therethrough.

The electrically-conductive plate **112**, **122** of each jaw member **110**, **120** defines a generally planar tissue-contacting surface **112a**, **122a**, an elongated slot **112b**, **122b** extending through the respective tissue-contacting surface **112a**, **122a**, a pair of legs **112c**, **122c** extending downwardly from each side of the respective tissue-contacting surface **112a**, **122b**, and a distal edge **112d**, **122d** disposed at the distal end of the respective tissue-contacting surface **112a**, **122a**. Electrically-conductive plates **112**, **122** extend from the proximal heels of jaw members **110**, **120**, e.g., the interface between flanges **113a**, **123a** and the distal portions of jaw members **110**, **120**, to the distal tips of jaw members **110**, **120**. Jaw housing **118** of jaw member **110** includes a pair of proximal tissue stops that extend therefrom about either side of jaw member **120** such that, in conjunction with the positioning of electrically-conductive plates **112**, **122** at the proximal heel of jaw members **110**, **120**, grasping of tissue proximally of electrically-conductive plates **112**, **122** is inhibited.

Tissue-contacting surfaces **112a**, **122a** define a plurality of spaced-apart recesses **112e**, **122e** therein that facilitate grasping tissue. Tissue-contacting surface **112a** of electrically-conductive plate **112** of jaw member **110** and/or tissue-contacting surface **122a** of electrically-conductive plate **122** of jaw member **120** may further include a plurality of stop members **122f** disposed thereon. Stop members **122f** may be constructed of a heat-resistant ceramic deposited onto the tissue-contacting surfaces **112a**, **122a**, an electrically non-conductive plastic molded onto tissue-contacting surfaces **112a**, **122a**, an electrical conductive material isolated from the respective tissue-contacting surface **112a**, **122a**, or may be formed from and/or manufactured in any other suitable fashion.

Each wing **116b** of spacer **115** of jaw member **110** defines a slot **116g**, open at the top end thereof, that is configured for receiving one of the legs **112c** of electrically-conductive plate **112**. Wire **107**, which extends through tunnel **116f** defined within spacer **115** is configured to electrical connect to an underside of electrically-conductive plate **112** towards the distal end thereof for enabling the selective supply of energy thereto. Wire **107** is configured to extend proximally through shaft **80** and into housing **20**, ultimately coupling to energy activation assembly **190** (FIG. 7) and/or extending through the cable (not shown) to couple to the generator (not shown).

Outer housings **118**, **128** are formed about jaw members **110**, **120** via a second overmold process, such that each outer housing **118**, **128** partially encloses respective jaw members **110**, **120** with the exception of a portion of the distal jaw support **113b**, **123b** thereof and the tissue-contacting surface **112a**, **122a** thereof, which remain exposed. Further, legs **112c**, **122c** of electrically-conductive plates **112**, **122** of jaw members **110**, **120** and the spacers **115** (only spacer **115** of

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jaw member **110** is shown) thereof each define a plurality of fill-apertures **122g** (only fill-apertures **122g** of electrically-conductive plate **122** of jaw member **120** are illustrated) that, upon overmolding of outer housings **118**, **128** about respective jaw members **110**, **120** are filled with the overmolded material forming outer housings **118**, **128** to lock the components of each jaw member **110**, **120** in an assembled condition. Further, outer housings **118**, **128** define lengths extending along the sides of respective jaw members **110**, **120** and thicknesses that decrease in the proximal-to-distal direction along the lengths thereof. Outer housings **118**, **128** also define windows **119**, **129** that align with and communicate with the windows **116d** of the respective spacers **115** (only spacer **115** of jaw member **110** is illustrated) and the knife slots **112b**, **122b** of the respective electrically-conductive plate **112**, **122** thereof so as to define an opening **131**, **132** extending through the distal portion of each jaw member **110**, **120** transversely relative to the plane defined by the respective tissue-contacting surface **112a**, **122a**.

With outer housings **118**, **128** formed about jaw members **110**, **120**, respectively, distal edges **112d**, **122d** of electrically-conductive plates **112**, **122** overlap the distal ends of outer housings **118**, **128** such that, as illustrated in FIGS. **15** and **16**, distal edges **112d**, **122d** can be utilized to pinch tissue therebetween. In particular, this configuration enables pinching of planar tissue structures that lack substantial protruding portions that would otherwise enable grasping, such as the tissue wall illustrated in FIGS. **15** and **16**.

Referring to FIGS. **15A** and **16A**, jaw members **110**, **120** may further be utilized to spread and/or dissect tissue. In order to do so, with jaw members **110**, **120** disposed in the approximated position, end effector assembly **100** may be manipulated such that the distal tips of jaw members **110**, **120** are pressed into contact with tissue to be spread and/or dissected, as shown in FIG. **15A**. Thereafter, jaw members **110**, **120** are moved to the spaced-apart position such that the distal ends of outer housings **118**, **128** of jaw members **110**, **120**, respectively, push tissue in opposite directions, thus spreading and/or dissecting tissue. Various configurations of the distal ends of outer housings **118**, **128** of jaw members **110**, **120** to further facilitate spreading and/or dissecting tissue are detailed below. As also detailed below, drive assembly **140** (FIG. **12**) defines a pre-loaded configuration wherein drive assembly **140** (FIG. **12**) is always under tension, such that backlash upon moving jaw members **110**, **120** from the approximated position back to the spaced-apart position is eliminated. Such a configuration facilitates spreading and/or dissecting tissue by allowing for a more smooth and consistent transition of jaw members **110**, **120** from the approximated position back to the spaced-apart position.

Turning to FIGS. **15B**, **15B'**, and **16B**, in some embodiments, the distal ends of outer housings **118**, **128** of jaw members **110**, **120** define cut-outs that form shelves **118b**, **128b** between the distal ends of outer housings **118**, **128** and the body portions **118a**, **128a** of housings **118**, **128**, respectively. Shelves **118b**, **128b**, as shown in FIGS. **15B**, **15B'**, and **16B** facilitate the retention of tissue via the distal ends of outer housings **118**, **128**, thus inhibiting slipping of tissue and facilitating spreading and/or dissecting tissue.

Referring to FIGS. **15C**, **15C'**, and **16C**, in some embodiments, in addition to or as an alternative to including shelves **118b**, **128b**, the distal ends of outer housings **118**, **128** of jaw members **110**, **120** define extensions **118c**, **128c** that are relatively narrow and relatively small-radiused as compared to body portions **118a**, **128a** of housings **118**, **128**. These extensions **118c**, **128c** facilitate pressing the distal ends of

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jaw members 110, 120 further into tissue (see FIG. 15C) to ensure a relatively large contact area of tissue against shelves 118b, 128b upon moving jaw members 110, 120 to the spaced-apart position (see FIG. 16C), thus facilitating the spreading and/or dissecting of tissue.

With reference to FIGS. 15D, 15D', and 16D, in some embodiments, the distal ends of outer housings 118, 128 of jaw members 110, 120 are cut-back to define angled surfaces 118d, 128d that define an angle 1° relative to the perpendicular extending from the distal ends of jaw members 110, 120 (see FIG. 16D). Angled surfaces 118d, 128d, similarly as with the previous embodiments, facilitate the pressing of the distal ends of jaw members 110, 120 further into tissue (see FIG. 15D) as well as the retention of tissue while spreading and/or dissecting tissue.

Referring again to FIGS. 7-12, trigger assembly 70, as mentioned above, is operably coupled to knife assembly 170 to enable selective translation of knife blade 174 of knife assembly 170 relative to end effector assembly 100. Trigger assembly 70 includes trigger 72 and a linkage 76. Trigger 72 includes a grasping portion 75a which includes the concave trigger surface 73, a pivot extension 75b extending upwardly from grasping portion 75a, and a proximal extension 75c extending proximally from grasping portion 75a. Grasping portion 75a also includes a tab 75d extending distally therefrom. Tab 75d defines an aperture 75e configured to retain movable end 71b of biasing member 71 therein. As noted above, fixed end 71a of biasing member 71 is engaged via retention pin 29f of first housing component 22a of housing 20. In this manner, biasing member 71 serves to bias trigger 72 distally towards an un-actuated position (FIG. 35).

Pivot extension 75b of trigger 72 is pivotably coupled to housing 20 via pivot pin 48, which is engaged within and extends between pivot apertures 29c of first and second housing components 22a, 22b of housing 20. It is noted that pivot pin 48 is shared by both trigger 72 and movable handle 40; that is, both trigger 72 and movable handle 40 are pivotable about the same point relative to housing 20. Proximal extension 75c of trigger 72 includes a post 75f that, as detailed below, is operably engaged within cam slot 77e of linkage 76.

Linkage 76 serves to operably couple trigger 72 with knife assembly 170 such that pivoting of trigger 72 from the un-actuated position (FIG. 35) to the actuated position (FIG. 36) advances knife blade 174 relative to end effector assembly 100 to cut tissue grasped between jaw members 110, 120, as detailed below. Linkage 76 defines a generally triangular-shaped configuration including an apex 77a pointing in a distal direction and a base defining upper and lower corners 77b, 77c, respectively. Apex 77a includes a peg 77d that is configured for receipt within pivot boss 29e of first housing component 22a to pivotably couple linkage 76 relative to housing 20 about apex 77a thereof. A cam slot 77e, 77f is defined through linkage 76 adjacent upper and lower corners 77b, 77c, respectively. A coupling pin 78 operably couples cam slot 77e with knife plate 172 of knife assembly 170. More specifically, coupling pin 78 includes a cap 79a defining a slot 79c (FIG. 9) configured to receive finger 173 of knife plate 172 and a rod 79b that is operably engaged within cam slot 77e. As noted above, post 75f of proximal extension 75c of trigger 72 is operably engaged within cam slot 77f.

As a result of the above-detailed configuration of trigger assembly 70, pivoting of trigger 72 between the un-actuated and actuated positions (FIGS. 35 and 36, respectively) urges linkage 76 to pivot relative to housing 20 ultimately such that coupling pin 78 is urged to translate longitudinally

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within and relative to housing 20. As finger 173 of knife plate 172 is engaged with coupling pin 78, such longitudinal translation of coupling pin 78 is imparted to knife plate 172 for translating knife blade 174 between retracted and extended positions (FIGS. 29-34 and 38-40, respectively) relative to end effector assembly 100, as detailed below.

Knife assembly 170, as noted above, includes a knife plate 172 defining a finger 173 at the proximal end thereof. Knife plate 172 extends distally through housing 20 and shaft 80 atop drive plate 142 and is slidably engaged therewith via receipt of each end of knife plate 172 within track edges 146 of drive plate 142. Knife assembly 170 further includes knife blade 174 integrally formed with or otherwise engaged to knife plate 172 and extending distally therefrom. Knife blade 174 defines a width less than the combined thickness of jaw members 110, 120 at the proximal ends thereof but greater than or equal to the combined thickness of jaw members 110, 120 at the distal ends thereof. Knife blade 174 further defines an elongated opening 176 extending longitudinally therethrough. Elongated opening 176 permits knife blade 174 to be slidably disposed about pivot pin 103 and cam pin 105. More specifically, elongated opening 176 defines a first portion 177a having a first width configured to slidably receive pivot pin 103 and a second portion 177b having a second width configured to slidably receive cam pin 105 but sufficiently small to inhibit receipt of the larger-diameter pivot pin 103 therein.

As appreciated in view of the above, handle assembly 30, slider assembly 150 of drive assembly 1450, and trigger assembly 70 enable efficient assembly of instrument 10 in that these components may be operably positioned within housing 20 and relative to one another via a top-down assembly process.

Turning now to FIGS. 22-40, the use and operation of instrument 10 is described. Initially, as illustrated in FIGS. 22-24, 29, 31, and 33, movable handle 40 is disposed in the initial position and, correspondingly, jaw members 110, 120 are disposed in the spaced-apart position. More specifically, with movable handle 40 in the initial position, engagement bulge 51 is disposed in a distal-most position such that slider assembly 150 is disposed in a distal-most position. With slider assembly 150 disposed in its distal-most position, torsion spring 160 is less-tensioned and second leg 162 of torsion spring 160 retains drive plate 142 in a distal-most position. Torsion spring 160 is less-tensioned but not fully un-tensioned in the initial position of movable handle 40. This configuration maintains a pre-load on drive assembly 140 such that, as noted above, backlash due to the complete removal of tension from torsion spring 160 as jaw members 110, 120 move from the approximated position back to the spaced-apart position is eliminated. Drive plate 142 is inhibited from moving proximally relative to slider assembly 150 in this position due to the abutment of the proximal edge 145 of drive plate 142 with abutment rib 153d of proximal housing 152 of slider assembly 150. Further, in this distal-most position of drive plate 142, drive plate 142 maintains cam pin 105 at the distal ends of cam slots 114b, 124b and, thus, jaw members 110, 120 are maintained in the spaced-apart position.

Turning for the moment to FIG. 22A, in some embodiments, rather than first leg 161 of torsion spring 160 (FIGS. 10-11) being fixed relative to proximal housing 152 of slider assembly 150, a torsion spring 3160 may be provided including a first leg 3161 that extends through a slot 3153b defined within proximal housing 152 and is positioned to abut a block 3020 mounted within or monolithically formed with housing 20. As a result of this configuration, torsion

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spring 3160 provides the additional function of biasing slider assembly 150 distally, thereby biasing jaw members 110, 120 towards the spaced-apart position (see FIG. 14). That is, upon movement of movable handle 40 to translate slider assembly 150 proximally (see FIG. 9), first leg 3161 of torsion spring 3160 is maintained in position via its abutment with block 3020, thereby further tensioning torsion spring 3160 such that, upon release of movable handle 40 (FIG. 9), torsion spring 3160 serves to bias slider assembly 150 distally, thereby biasing jaw members 110, 120 towards the spaced-apart position (see FIG. 14). Further, multiple blocks 3020 may be provided at different positions within housing 20 such that, during assembly, first leg 3161 of torsion spring 3160 may be positioned to abut a selected one of the blocks 3020 to achieve a desired biasing force.

Returning to FIGS. 22-24, 29, 31, and 33, trigger 72 is initially disposed in the un-actuated position and, accordingly, knife blade 174 is disposed in the retracted position. More specifically, in the un-actuated position, trigger 72 is disposed in a distal-most position under the bias of biasing member 71 such that lower corner 77c of linkage 76, which is coupled to trigger 72 via engagement of post 75f within slot 77f, is disposed in a distal-most position. Since upper and lower corners 77b, 77c of linkage 76 are disposed on opposite sides of apex 77a, with lower corner 77c disposed in its distal-most position, upper corner 77b is disposed in a proximal-most position. With upper corner 77b disposed in its proximal-most position, knife plate 172 is likewise disposed in a proximal-most position due to the engagement of pin 78 within slot 77e. The proximal-most position of knife plate 172 corresponds to the retracted position of knife blade 174, wherein knife blade 174 is disposed between flanges 113a, 123a of jaw members 110, 120 but does not extend distally therefrom so as to avoid interference with tissue disposed between jaw members 110, 120. Further, in this position, pivot pin 103 is disposed at the distal end of first portion 177a of opening 176 of knife blade 174 and cam pin 105 is likewise disposed within first portion 177a of opening 176.

With additional reference to FIGS. 25-28, 30, 32, and 34, in order to move jaw members 110, 120 to the approximated position to grasp tissue therebetween, movable handle 40 is pulled proximally towards fixed handle portion 26 from the initial position (FIG. 22) to the compressed position (FIG. 27). Upon such movement of movable handle 40 to the compressed position, engagement bulge 51 of movable handle 40 moves proximally relative to housing 20, thereby urging slider assembly 150 proximally through housing 20. Torsion spring 160, in the less-tensioned state, is translated proximally together with slider assembly 150 such that second leg 162 of torsion spring 160 pulls drive plate 142 proximally in connection with the proximal translation of slider assembly 150. In other words, at this point, slider assembly 150 and drive plate 142 move in concert with one another. As drive plate 142 is pulled proximally, cam pin 105 is pulled proximally through cam slots 114b, 124b such that jaw members 110, 120 are pivoted from the spaced-apart position to the approximated position to grasp tissue therebetween.

As detailed above, movement of movable handle 40 from the initial position (FIG. 22) to the compressed position (FIG. 27) similarly translates drive plate 142 proximally, thereby moving jaw members 110, 120 to the approximated position to grasp tissue therebetween. Drive plate 142 is still inhibited from moving proximally relative to slider assembly 150 in this position due to the abutment of the proximal

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edge 145 of drive plate 142 with abutment rib 153d of proximal housing 152 of slider assembly 150.

At this point, with tissue grasped between jaw members 110, 120, instrument 10 may be utilized as a pliers to maneuver, manipulate, and/or reposition tissue. In particular, as noted above and illustrated in FIGS. 15 and 16, jaw members 110, 120 may be approximated adjacent a wall of tissue to pinch tissue between the distal tips of jaw members 110, 120 to enable maneuvering, manipulating, and/or repositioning thereof.

Referring to FIG. 35, in order to apply energy to tissue grasped between jaw members 110, 120 to treat tissue, movable handle 40 is compressed further towards fixed handle portion 26 to an activation position, wherein an appropriate closure force or closure force within an appropriate range, is achieved and energy activation is initiated. As movable handle 40 is moved further proximally relative to housing 20 beyond the compressed position, an appropriate closure force or closure force within an appropriate range is imparted to tissue grasped between electrically-conductive plates 112, 122 of jaw members 110, 120 regardless of the thickness or compressibility of tissue or the position of movable handle 40. This is because, upon movement of movable handle 40 from the compressed position towards the activation position, slider assembly 150 is translated proximally while drive plate 142 is maintained in position. In other words, upon movement of movable handle 40 from the compressed position to the activated position, slider assembly 150 and drive plate 142 no longer move in concert with one another. Rather, as detailed below, slider assembly 150 and drive plate 142 are decoupled to permit relative motion therebetween.

The decoupling of slider assembly 150 and drive plate 142 to permit relative motion therebetween is provided via torsion spring 160. More specifically, upon proximal movement of movable handle 40, a first force is imparted from movable handle 40, through slider assembly 150, first leg 161 of torsion spring 160, the body of torsion spring 160, and second leg 162 of torsion spring 160, to drive plate 142 to urge drive plate 142 in a proximal direction, while a second, opposite force acts on drive plate 142 and, thus, second leg 162 of torsion spring 160 in a distal direction to resist further compression of tissue between jaw members 110, 120. Once the second, opposite force exceeds the spring force of torsion spring 160, proximal movement of slider assembly 150 no longer results in proximal movement of drive plate 142 but, rather, results in further tensioning of torsion spring 160, which absorbs the force imparted thereto from movement of movable handle 40. Thus, once this point as been reached, further proximal translation of slider assembly 150 urges first end 161 of torsion spring 160 proximally, while second opposite force retains second leg 162 of torsion spring 160 in position, thereby further tensioning torsion spring 160. Since second leg 162 of torsion spring 160 is retained in position, drive plate 142 is likewise retained in position despite the proximal translation of movable handle 40 and slider assembly 150.

It is noted that, during movement of movable handle 40 from the initial position to the compressed position, as detailed above, the second, opposite force is less than the spring force of torsion spring 160 and, thus, slider assembly 150, first and second legs 161, 162 of torsion spring 160, and drive plate 142 move in conjunction with one another. Further, torsion spring 160 may be configured such that the second, opposite force exceeds the spring force of torsion spring 160 at a force corresponding to a closure pressure on tissue between 3 kg/cm² to 16 kg/cm², as it has been found

that closure forces within this range facilitate sealing of tissue grasped between jaw members **110**, **120**. However, other forces and/or force ranges are also contemplated, e.g., for treating tissue in other manners (coagulating, cauterizing, etc.).

Continuing with reference to FIG. **35**, upon achieving the activation position of movable handle **40**, button activation post **196** (FIG. **7**) of movable handle **40** contacts depressible button **192** sufficiently so as to depress depressible button **192** into fixed handle portion **26** to activate switch **194**. Switch **194**, as noted above, is disposed in electrical communication with the generator (not shown) and electrically-conductive plates **112**, **122** of jaw members **110**, **120**, respectively, such that activation of switch **194** initiates the supply of energy to electrically-conductive plates **112**, **122** to treat, e.g., coagulate, cauterize, and/or seal, tissue grasped therebetween.

Referring to FIGS. **36-40**, once tissue has been treated or where it is only desired to cut tissue, knife blade **174** may be advanced between jaw members **110**, **120** to cut tissue grasped therebetween. In order to advance knife blade **174** from the retracted position to the extended position, trigger **72** is pulled proximally against the bias of biasing member **71** from the un-actuated position to the actuated position. As trigger **72** is pulled proximally, post **75f** is pushed proximally to urge linkage **76** to pivot counter-clockwise (from the orientation illustrated in FIG. **36**) such that upper corner **77b** of linkage **76** is moved distally. Distal movement of upper corner **77b** urges pin **78** to translate distally due to the engagement of rod **79b** of pin **78** within slot **77e** of upper corner **77b** of linkage **76**. Distal translation of pin **78**, in turn, urges knife plate **172** distally due to the engagement of finger **173** of knife plate **172** within cap **79a** of pin **78**.

As detailed above, movement of trigger **72** from the un-actuated position to the actuated position urges knife plate **172** distally. More specifically, knife plate **172** is urged distally such that knife blade **174** is advanced distally from the retracted position to the extended position. As knife blade **174** is advanced distally, knife blade **174** enters the knife slots of jaw members **110**, **120** defined by the respective knife slots **112b**, **122b** of electrically-conductive plates **112**, **122** and the knife slots **116c** of the respective spacers **115** (only spacer **115** of jaw member **110** is illustrated). As can be appreciated, translation of knife blade **174** through the knife slots of jaw members **110**, **120** to the extended position thereof divides tissue grasped between jaw members **110**, **120**.

Due to the fact that knife blade **174** defines a width greater than or equal to the combined thickness of jaw members **110**, **120** at the distal ends thereof, as knife blade **174** is advanced distally through the knife slots, knife blade **174** may extend at least partially through windows **119**, **129** and openings **131**, **132** of jaw members **110**, **120**, depending upon the thickness of tissue grasped between jaw members **110**, **120** (see FIG. **39**). Further, as knife blade **174** is advanced distally, pivot pin **103** and cam pin **105** translate proximally along opening **176** eventually such that cam pin **105** extends through second portion **177b** of opening **176**. As pivot pin **103** is too large to extend into second portion **177b** of opening **176**, interference therebetween defines the distal-most extent of travel of knife blade **174**. However, other components of knife assembly **170** and/or trigger assembly **70** may additionally or alternatively inhibit the extension of knife blade **174**.

Upon release of trigger **72**, trigger **72** and knife plate **172** are returned proximally under the bias of biasing member **71** such that knife blade **174** is returned to the retracted posi-

tion. Thereafter, movable handle **40** may be returned to the initial position to release the treated and/or divided tissue.

The various embodiments disclosed herein may also be configured to work with robotic surgical systems and what is commonly referred to as "Telesurgery." Such systems employ various robotic elements to assist the surgeon and allow remote operation (or partial remote operation) of surgical instrumentation. Various robotic arms, gears, cams, pulleys, electric and mechanical motors, etc. may be employed for this purpose and may be designed with a robotic surgical system to assist the surgeon during the course of an operation or treatment. Such robotic systems may include remotely steerable systems, automatically flexible surgical systems, remotely flexible surgical systems, remotely articulating surgical systems, wireless surgical systems, modular or selectively configurable remotely operated surgical systems, etc.

The robotic surgical systems may be employed with one or more consoles that are next to the operating theater or located in a remote location. In this instance, one team of surgeons or nurses may prep the patient for surgery and configure the robotic surgical system with one or more of the instruments disclosed herein while another surgeon (or group of surgeons) remotely control the instruments via the robotic surgical system. As can be appreciated, a highly skilled surgeon may perform multiple operations in multiple locations without leaving his/her remote console which can be both economically advantageous and a benefit to the patient or a series of patients.

The robotic arms of the surgical system are typically coupled to a pair of master handles by a controller. The handles can be moved by the surgeon to produce a corresponding movement of the working ends of any type of surgical instrument (e.g., end effectors, graspers, knives, scissors, etc.) which may complement the use of one or more of the embodiments described herein. The movement of the master handles may be scaled so that the working ends have a corresponding movement that is different, smaller or larger, than the movement performed by the operating hands of the surgeon. The scale factor or gearing ratio may be adjustable so that the operator can control the resolution of the working ends of the surgical instrument(s).

The master handles may include various sensors to provide feedback to the surgeon relating to various tissue parameters or conditions, e.g., tissue resistance due to manipulation, cutting or otherwise treating, pressure by the instrument onto the tissue, tissue temperature, tissue impedance, etc. As can be appreciated, such sensors provide the surgeon with enhanced tactile feedback simulating actual operating conditions. The master handles may also include a variety of different actuators for delicate tissue manipulation or treatment further enhancing the surgeon's ability to mimic actual operating conditions.

From the foregoing and with reference to the various figure drawings, those skilled in the art will appreciate that certain modifications can also be made to the present disclosure without departing from the scope of the same. While several embodiments of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise. Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular embodiments. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

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What is claimed is:

1. An electrosurgical end effector assembly, comprising:
first and second jaw members configured to grasp tissue
therebetween, the first jaw member including:
a jaw frame defining a proximal portion and a distal
portion, the proximal portion of the jaw frame
including first and second spaced-apart flanges
defining aligned pivot apertures configured to
receive a pivot pin to pivotably couple the first and
second jaw members with one another and aligned
cam slots configured to receive a cam pin to drive
pivoting of the first jaw member relative to the
second jaw member, the distal portion of the jaw
frame defining an upright and a base, wherein the
upright and the base are substantially perpendicular
relative to one another;
an electrode configured to conduct energy to tissue, the
electrode including a leg extending in substantially
parallel orientation relative to the upright, the leg
defining a plurality of apertures spaced-apart along at
least a portion of a length of the leg; and
an insulative housing extending along the leg and
through the plurality of apertures of the leg to secure
the electrode relative to the insulative housing.
2. The electrosurgical end effector assembly according to
claim 1, wherein the first jaw member defines a tunnel
extending longitudinally along a portion of the leg and
wherein a lead wire extends through the tunnel and electri-
cally connects with the leg.
3. The electrosurgical end effector assembly according to
claim 1, further comprising a clevis operably receiving the
first and second spaced-apart flanges of the proximal portion
of the jaw frame.
4. The electrosurgical end effector assembly according to
claim 1, wherein the first and second jaw members are
configured to conduct electrosurgical energy through tissue
grasped therebetween.
5. The electrosurgical end effector assembly according to
claim 1, wherein the electrode of the first jaw member and
an electrode of the second jaw member are configured to
conduct electrosurgical energy through tissue grasped ther-
ebetween.
6. The electrosurgical end effector assembly according to
claim 1, wherein the electrode includes an exposed portion
that is exposed above the insulative housing.
7. The electrosurgical end effector assembly according to
claim 6, wherein the exposed portion of the electrode defines
a tissue contacting surface.
8. The electrosurgical end effector assembly according to
claim 1, wherein the jaw frame is stainless steel.
9. The electrosurgical end effector assembly according to
claim 1, wherein the insulative housing is a first overmold.
10. The electrosurgical end effector assembly according to
claim 9, wherein the first jaw member further includes a
second overmold.
11. An electrosurgical end effector assembly, comprising:
first and second jaw members configured to grasp tissue
therebetween, the first jaw member including:

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- a jaw frame defining a proximal portion and a distal
portion, the proximal portion of the jaw frame
including first and second spaced-apart flanges
defining aligned pivot apertures configured to
receive a pivot pin to pivotably couple the first and
second jaw members with one another and aligned
cam slots configured to receive a cam pin to drive
pivoting of the first jaw member relative to the
second jaw member, the distal portion of the jaw
frame defining an upright and a base, wherein the
upright and the base are substantially perpendicular
relative to one another;
- an electrode configured to conduct energy to tissue, the
electrode including a leg defining an upper end and
a lower end, the leg extending in substantially par-
allel orientation relative to the upright, the leg defin-
ing a plurality of apertures spaced-apart along at
least a portion of a length of the leg and disposed
between the upper and lower ends of the leg, wherein
the upper end of the leg is exposed; and
- an insulative spacer capturing the lower end of the leg
within the insulative spacer, the insulative spacer
disposed on the jaw frame and positioned to electri-
cally isolate the electrode from the jaw frame.
12. The electrosurgical end effector assembly according to
claim 11, wherein the insulative spacer defines a tunnel
extending longitudinally along a portion of the leg and
wherein a lead wire extends through the tunnel and electri-
cally connects with the leg.
13. The electrosurgical end effector assembly according to
claim 11, wherein the first and second jaw members are
configured to conduct electrosurgical energy through tissue
grasped therebetween.
14. The electrosurgical end effector assembly according to
claim 11, wherein the electrode of the first jaw member and
an electrode of the second jaw member are configured to
conduct electrosurgical energy through tissue grasped ther-
ebetween.
15. The electrosurgical end effector assembly according to
claim 11, wherein the electrode includes an exposed portion
that is exposed above the insulative spacer.
16. The electrosurgical end effector assembly according to
claim 11, wherein the upright and the base of the jaw frame
define an interior angle side of the jaw frame and an exterior
angle side of the jaw frame, and wherein the insulative
spacer is at least partially disposed on the interior angle side
of the jaw frame.
17. The electrosurgical end effector assembly according to
claim 16, further comprising an insulative housing disposed
on the exterior angle side of the jaw frame.
18. The electrosurgical end effector assembly according to
claim 11, wherein the insulative spacer is a first overmold.
19. The electrosurgical end effector assembly according to
claim 18, wherein the first jaw member further includes a
second overmold.
20. The electrosurgical end effector assembly according to
claim 19, wherein the second overmold is an insulative
housing.

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